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Gerami et al.

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(54) **SEMI-STATIC HARQ CODEBOOK FOR DL-SPS**

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H04W 72/11 (2023.01)
(Continued)

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CPC **H04L 1/1812** (2013.01); **H04W 72/11** (2023.01); **H04W 72/1273** (2013.01); **H04W 72/21** (2023.01)

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See application file for complete search history.

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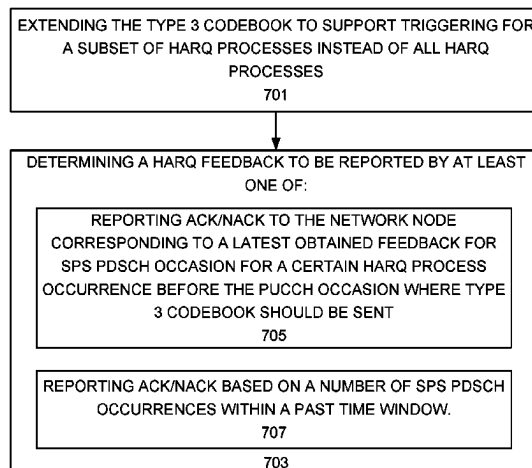
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(57) **ABSTRACT**

A method and communication device for reporting HARQ-ACK code books of fixed sizes in a communication network is provided. The method includes responsive to operating in a first mode, reporting the HARQ-ACK codebook for each scheduled transmission to a network node. The method further includes responsive to operating in a second mode, reporting the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is ACK, wherein no NACK is transmitted in the codebook when operating in the second mode. The method further includes responsive to operating in a third mode, reporting the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is NACK, wherein no ACK is transmitted in the codebook when operating in the second mode.

18 Claims, 12 Drawing Sheets



- (51) **Int. Cl.**
H04W 72/1273 (2023.01)
H04W 72/21 (2023.01)

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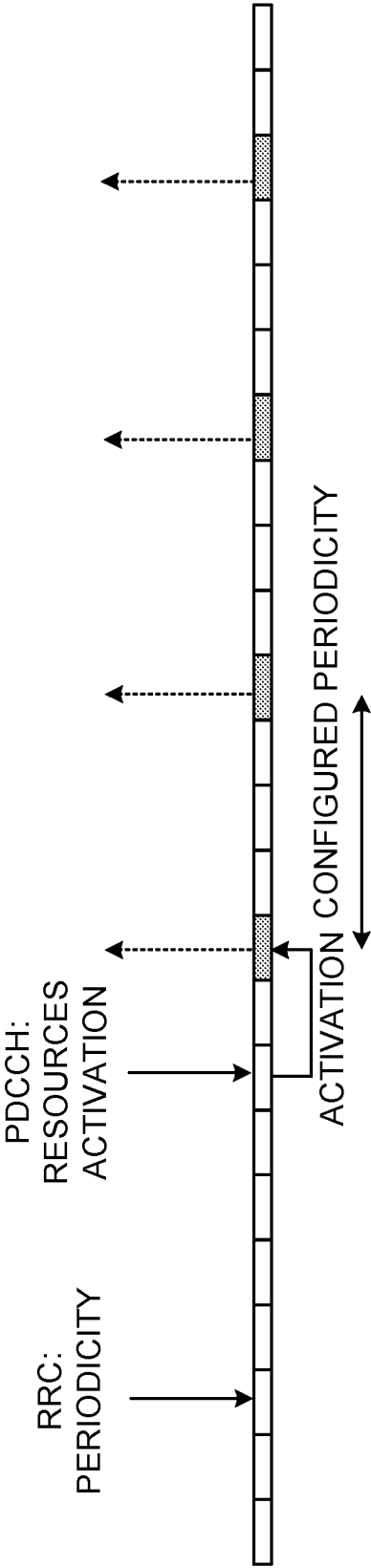


Figure 1

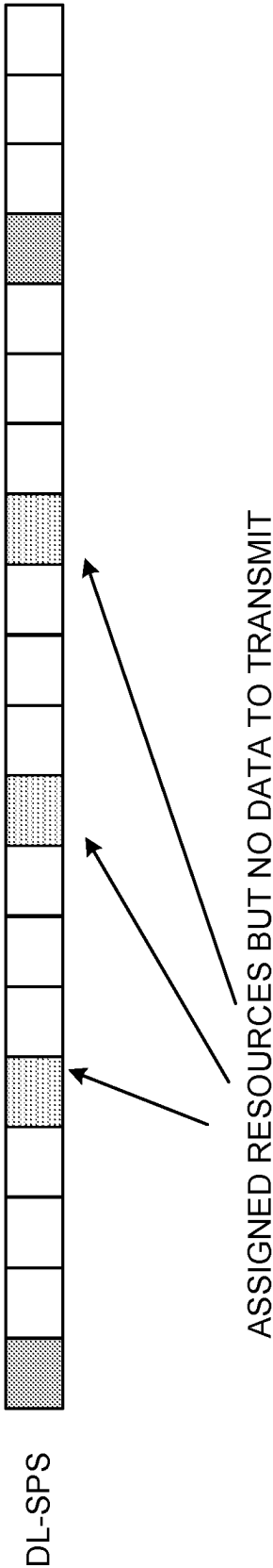
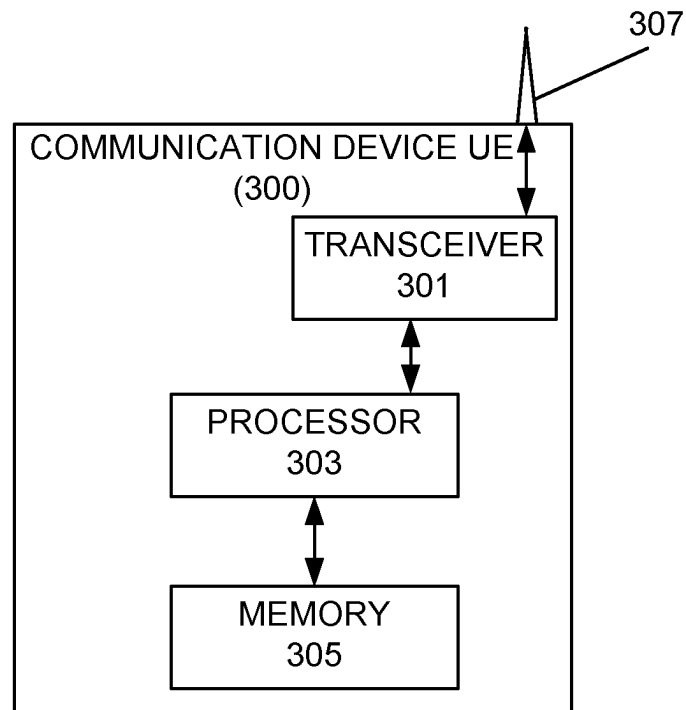
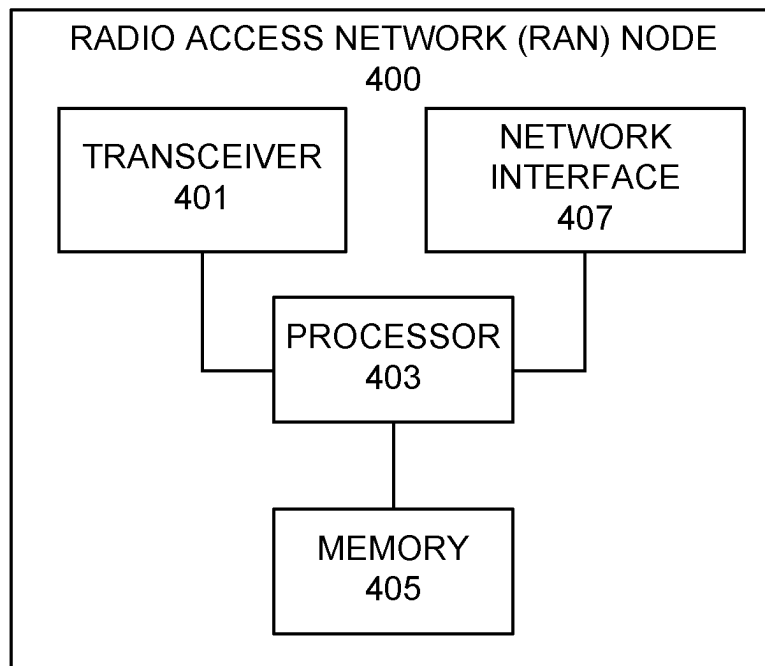
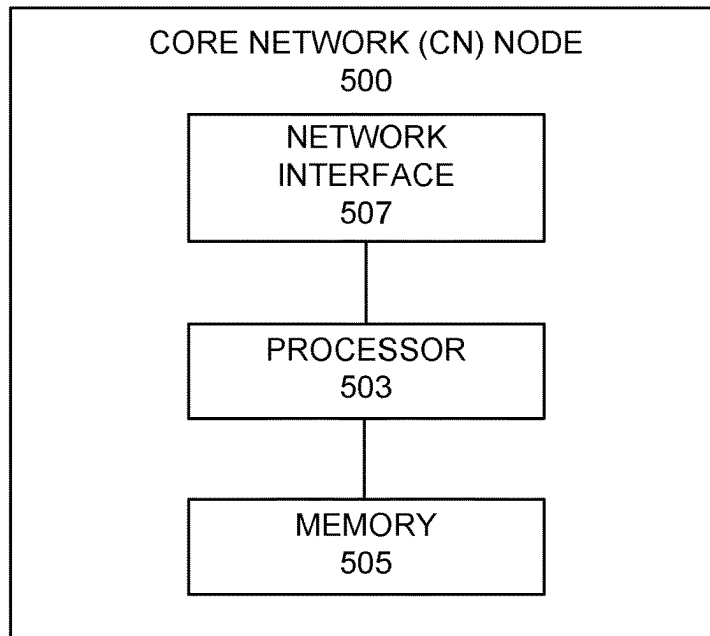
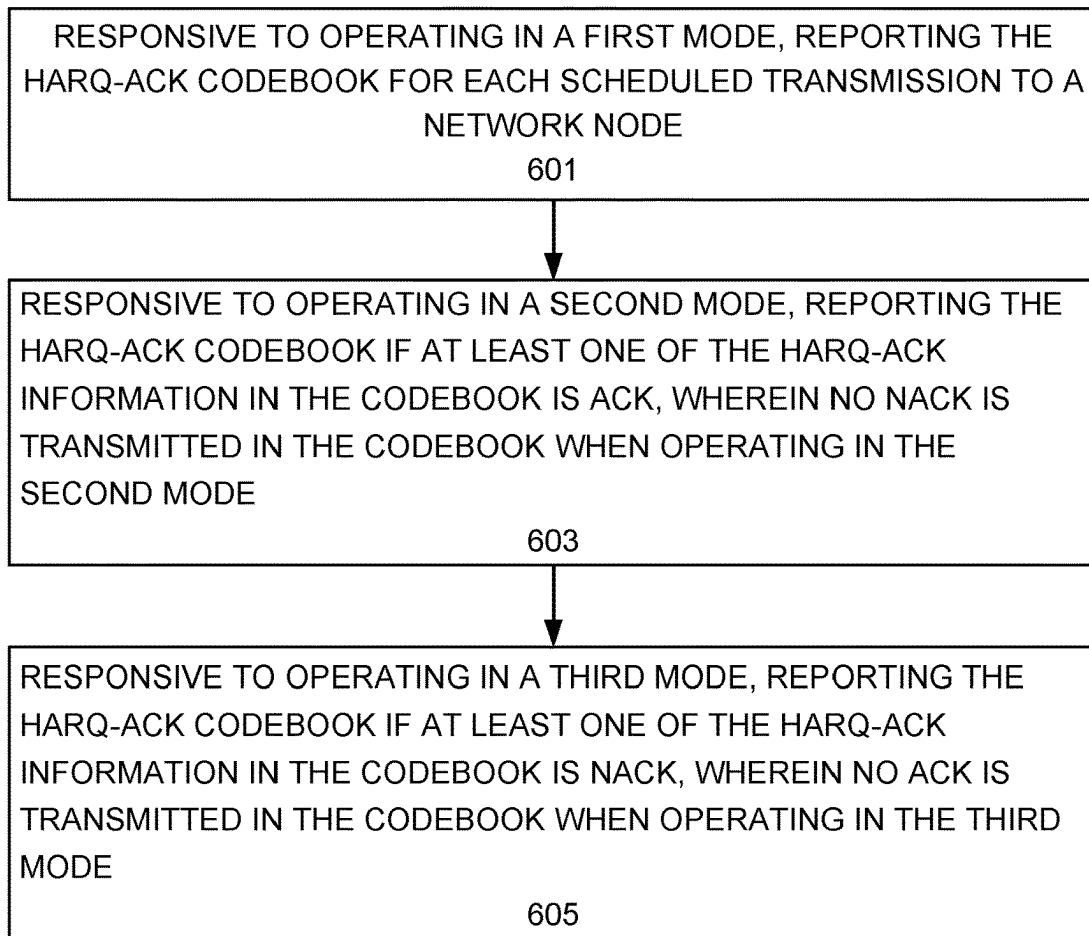
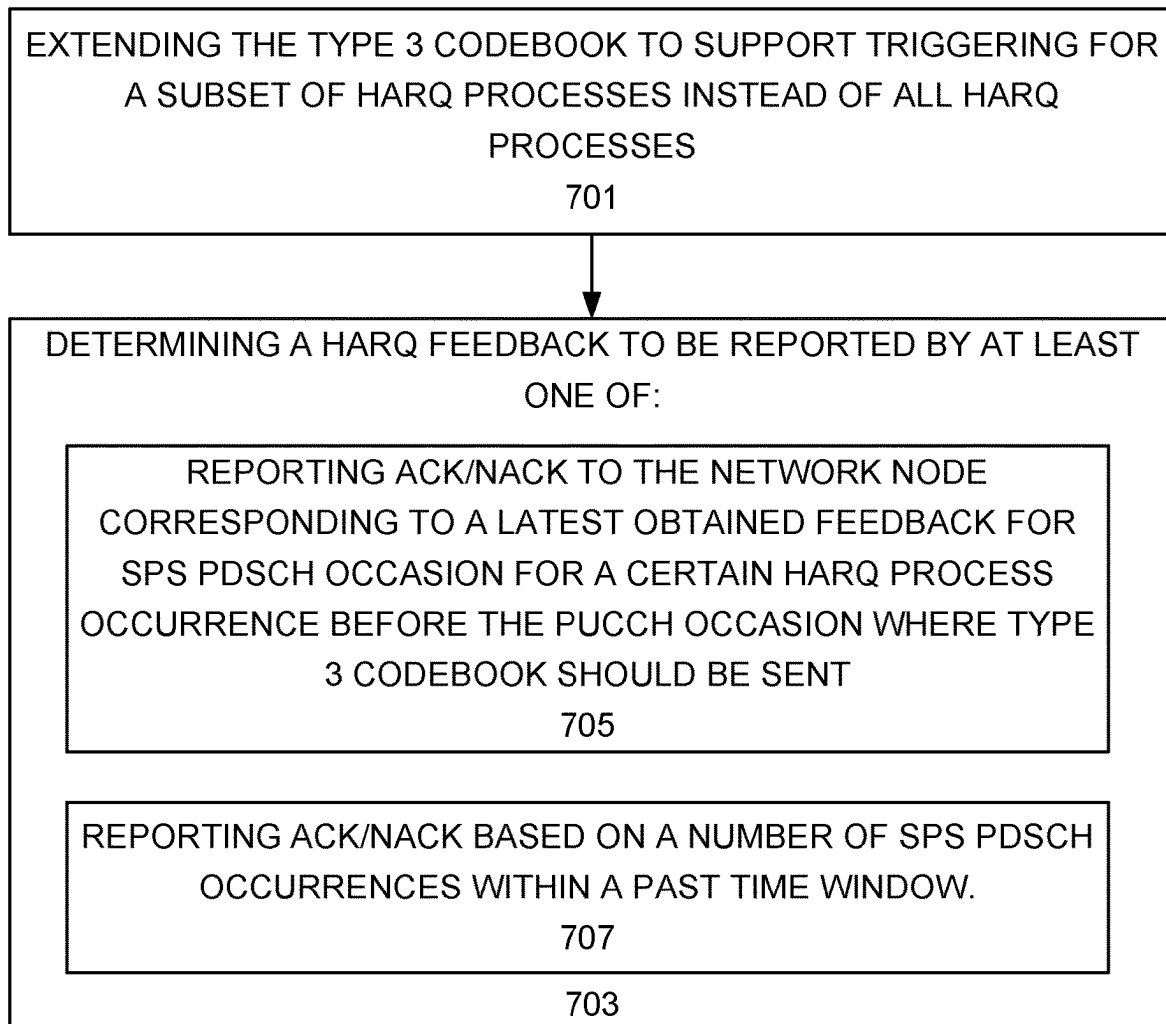


Figure 2

**Figure 3****Figure 4**

**Figure 5****Figure 6**

**Figure 7**

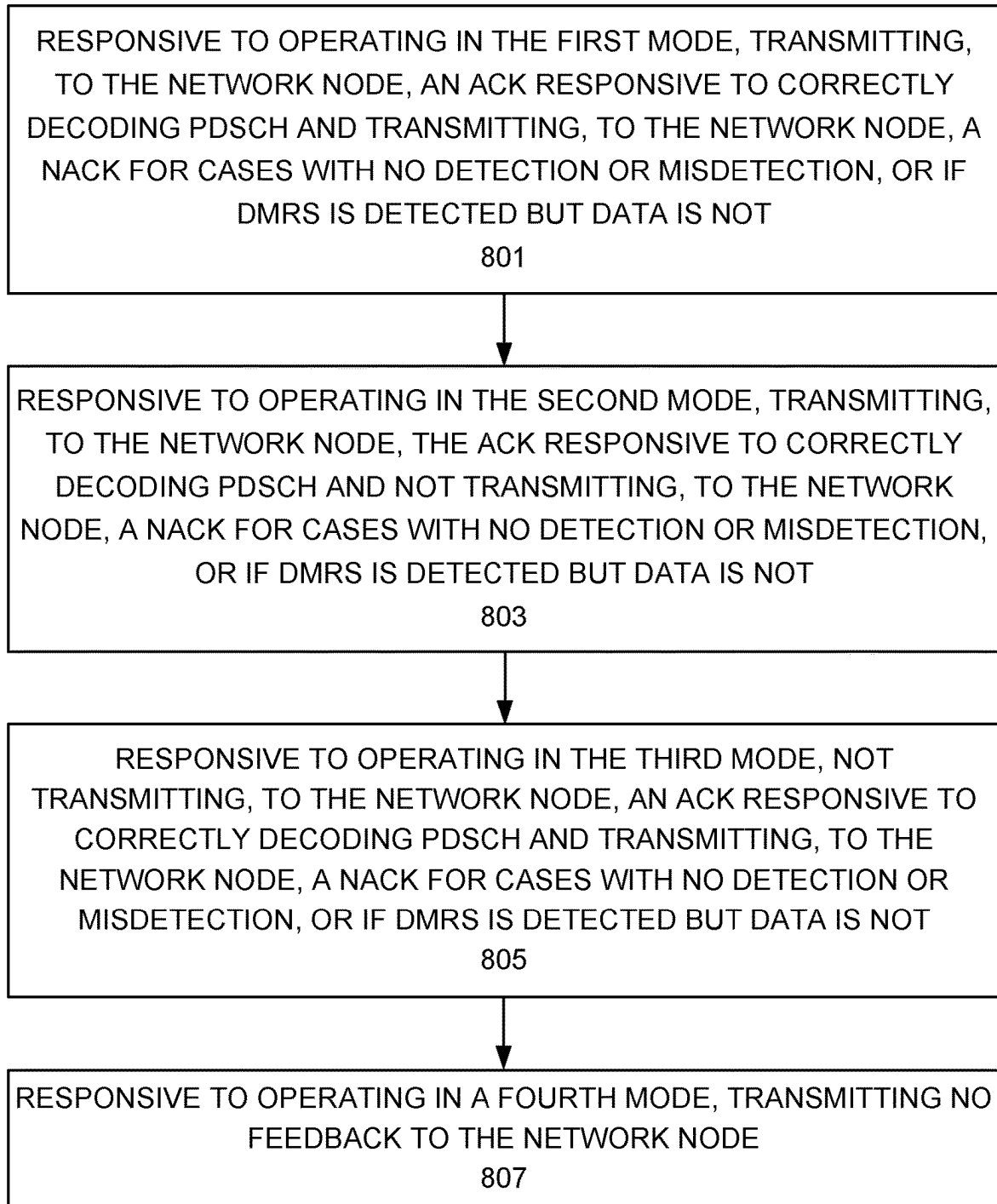
**Figure 8**

Figure 9

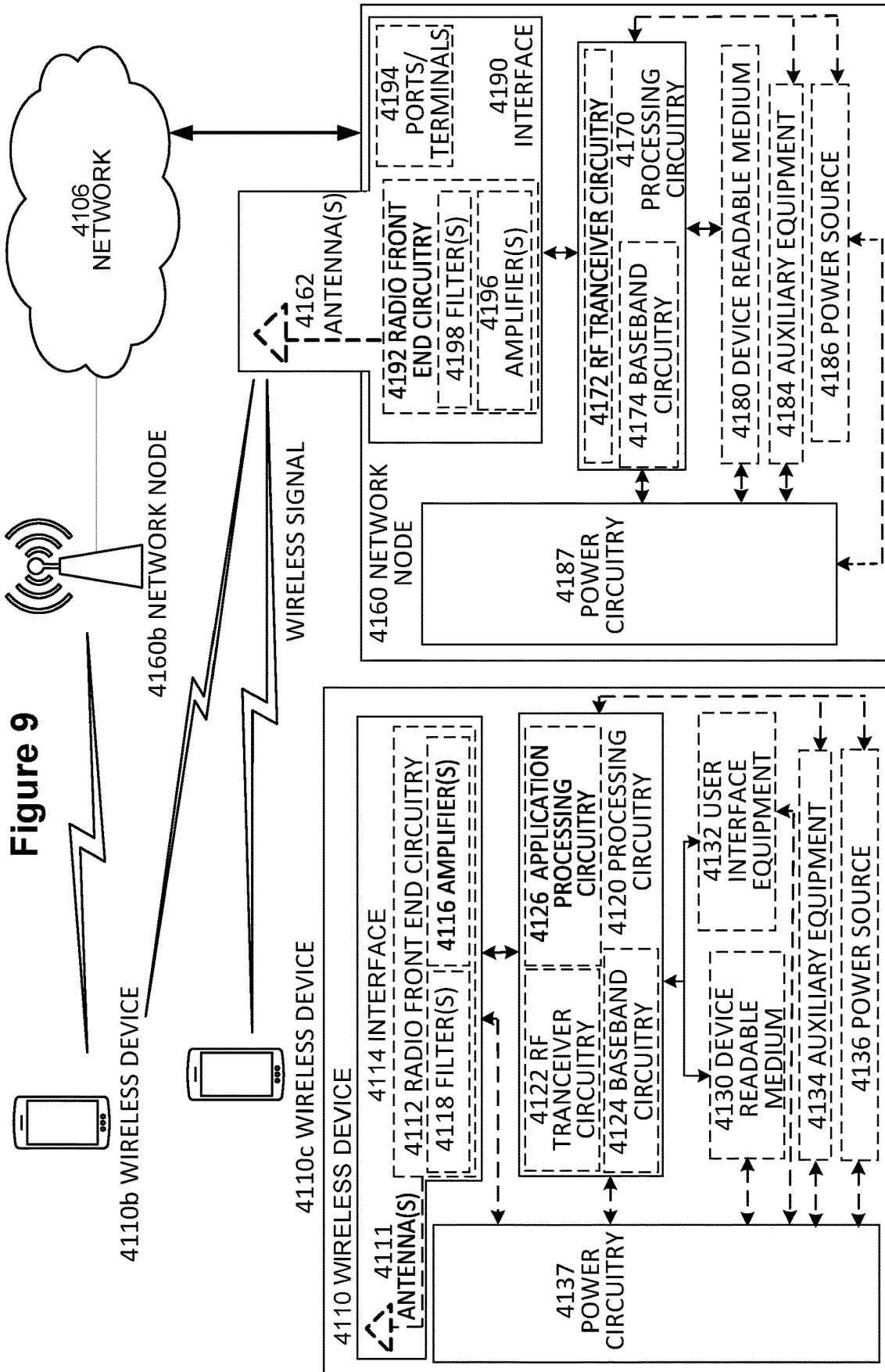


Figure 10

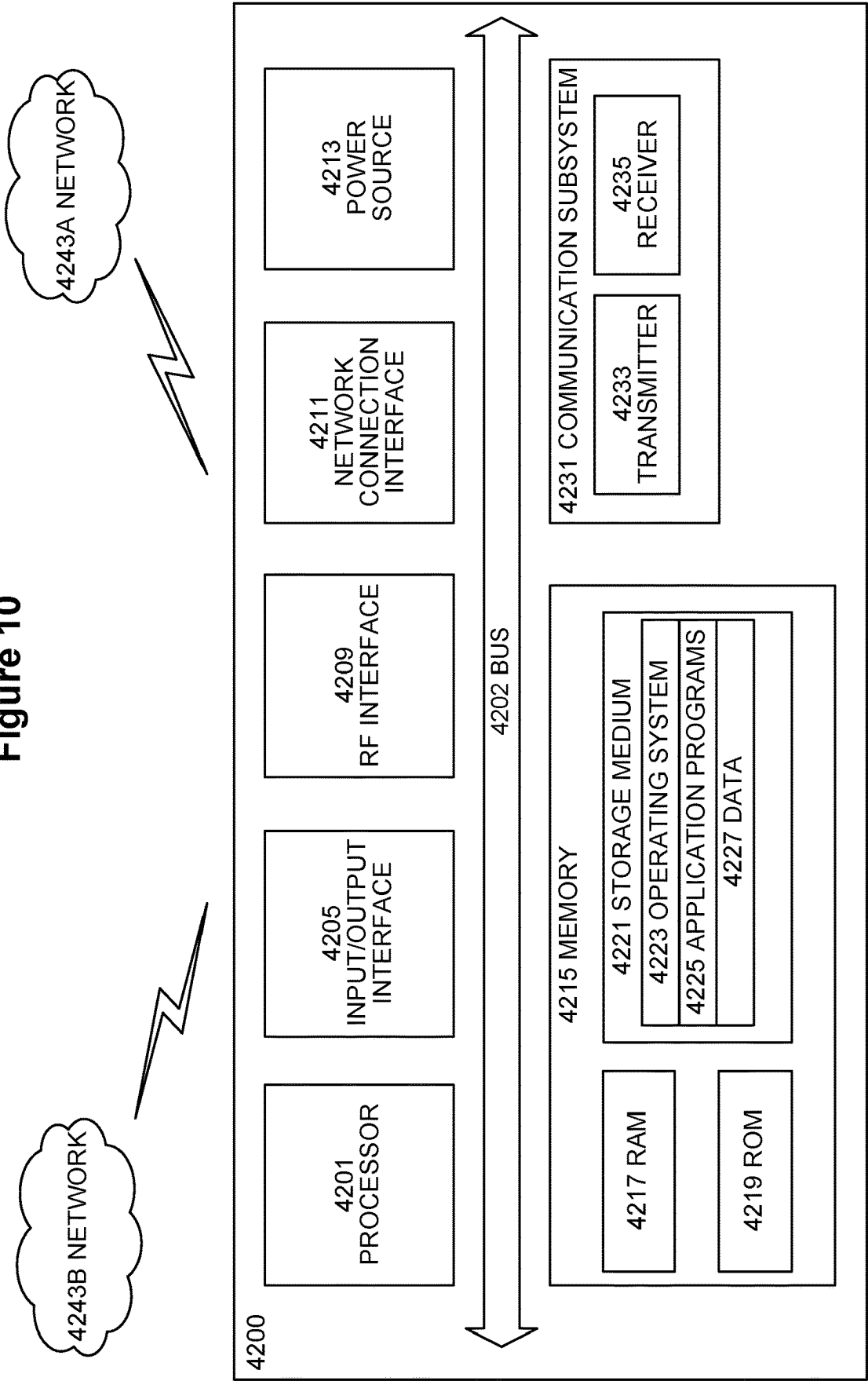
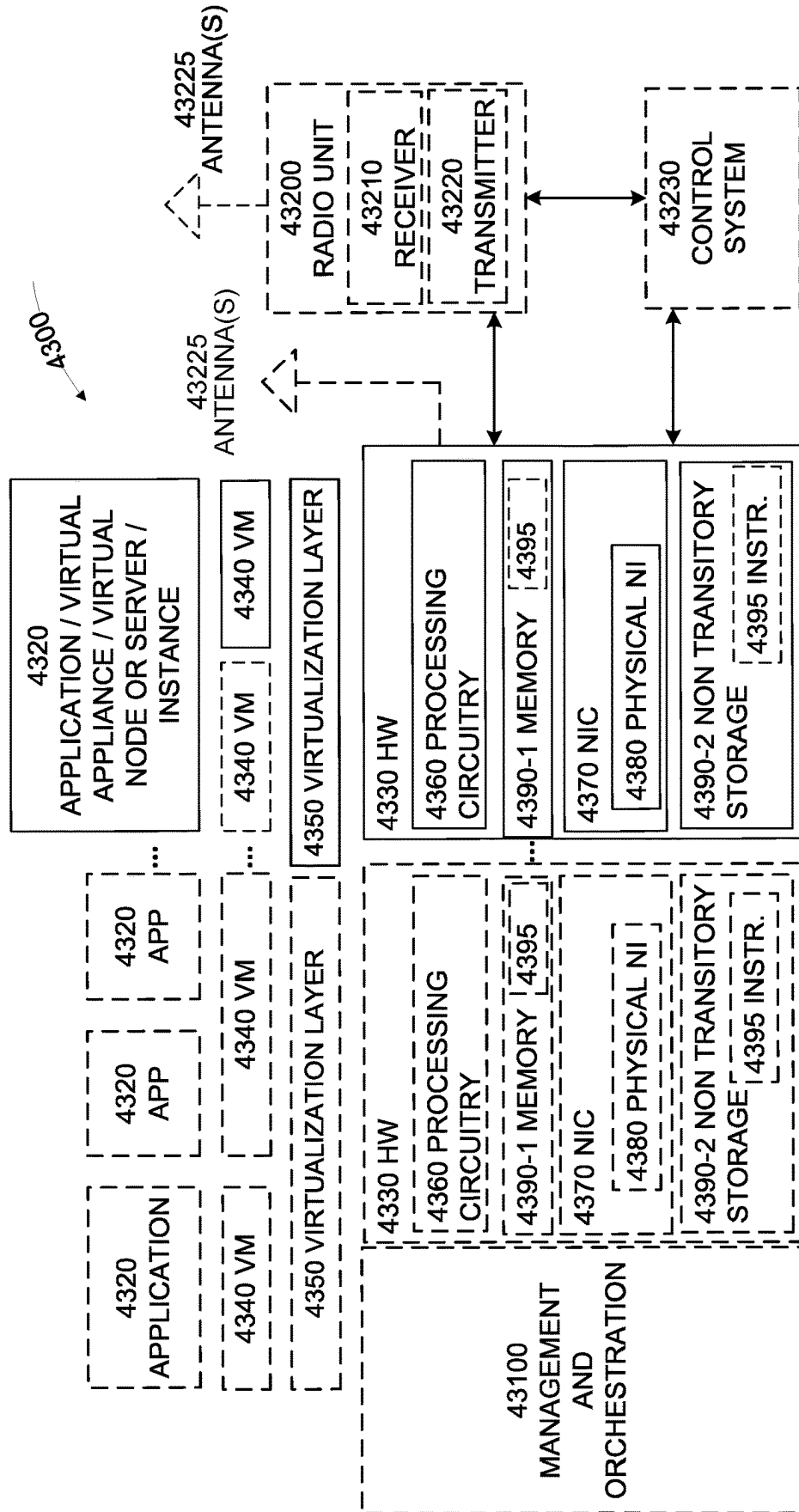


Figure 11



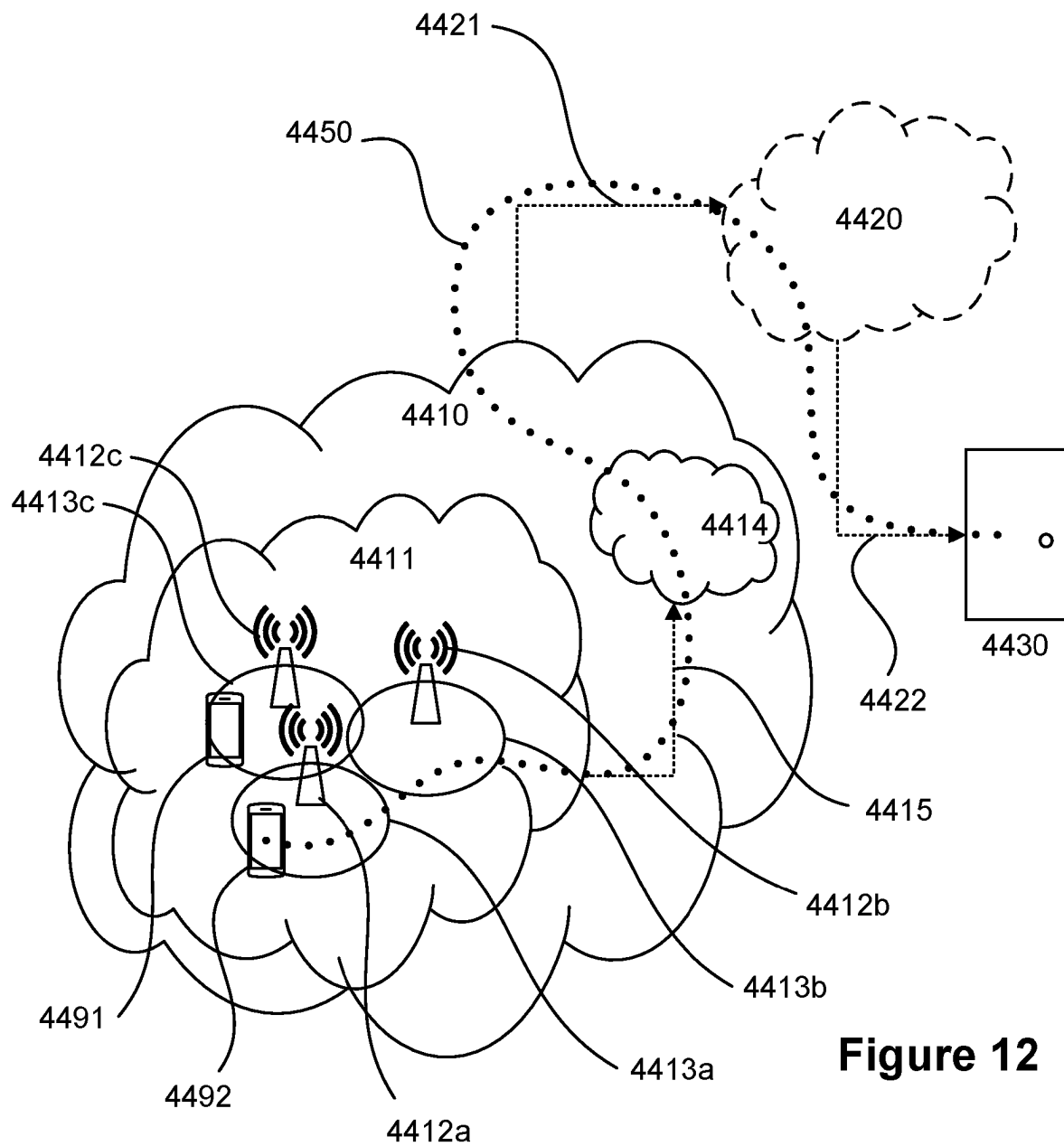
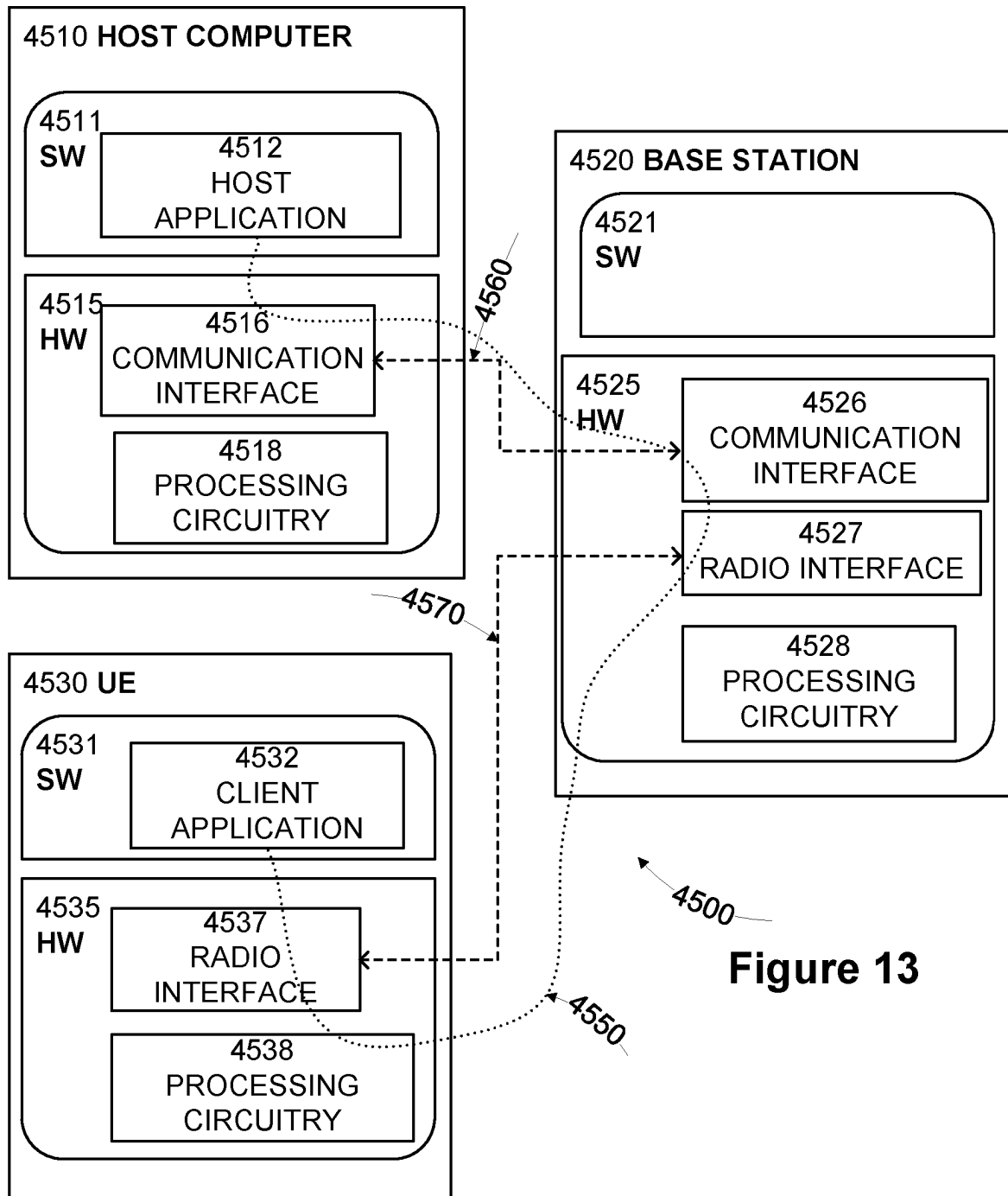
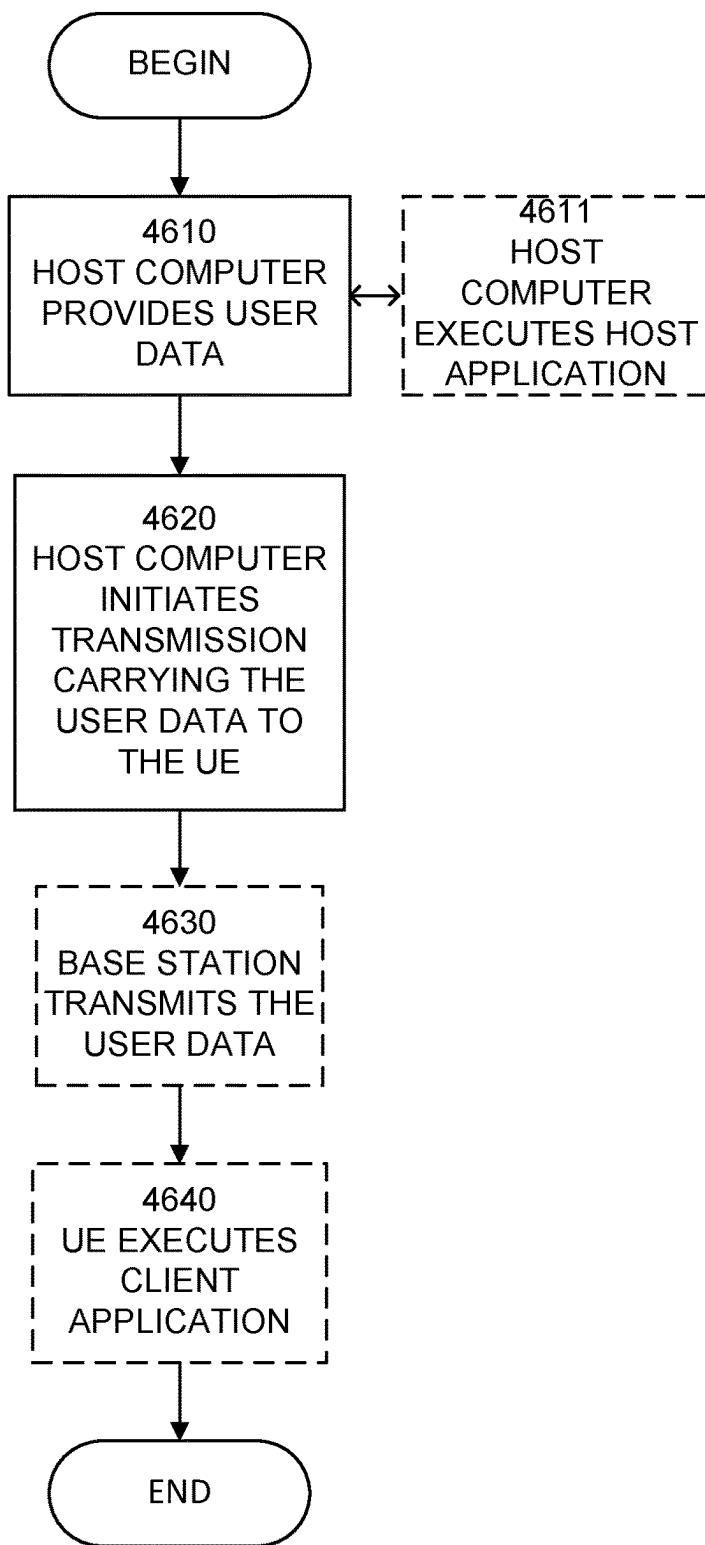
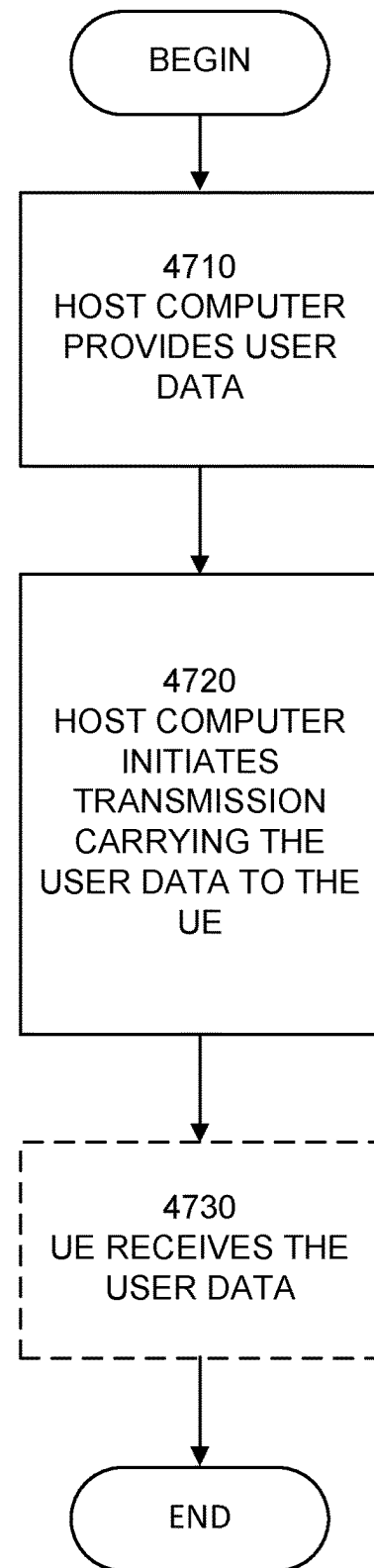


Figure 12

**Figure 13**

**Figure 14****Figure 15**

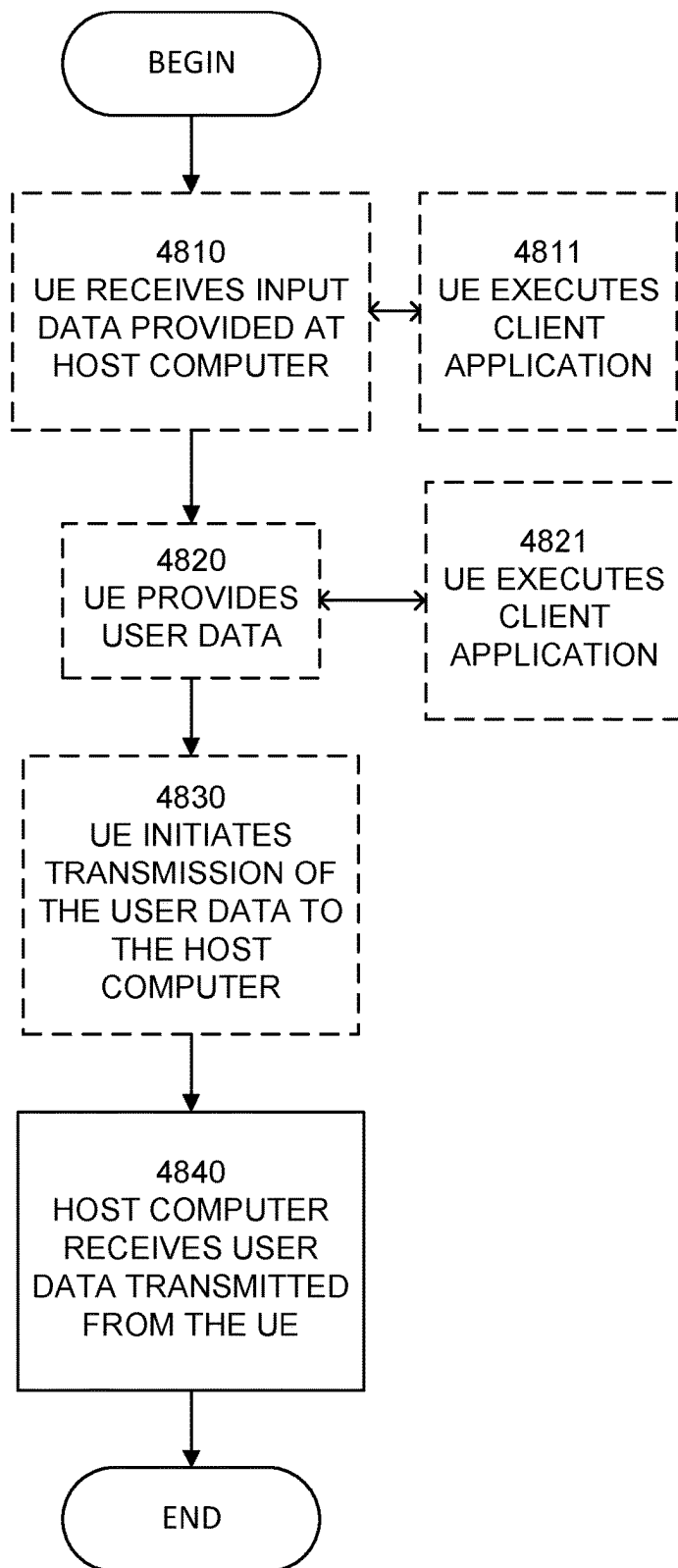


Figure 16

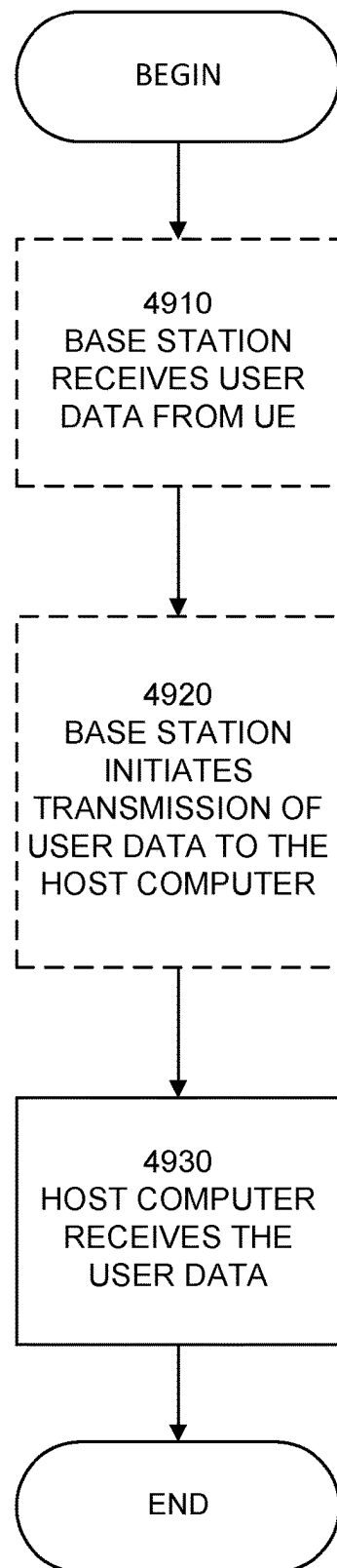


Figure 17

SEMI-STATIC HARQ CODEBOOK FOR DL-SPS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Submission Under 35 U.S.C. § 371 for U.S. National Stage Patent Application of International Application No.: PCT/SE2021/050354, filed Apr. 16, 2021 entitled “SEMI-STATIC HARQ CODEBOOK FOR DL-SPS,” which claims priority to U.S. Provisional Application No. 63/011,943, filed Apr. 17, 2020, entitled “SEMI-STATIC HARQ CODEBOOK FOR DL-SPS,” the entireties of both of which are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates generally to communications, and more particularly to communication methods and related devices and nodes supporting wireless communications.

BACKGROUND

A brief description of downlink semi-persistent scheduling (DL-SPS) [1]

DL-SPS is a scheme similar to semipersistent scheduling in long term evolution (LTE). A semi-static scheduling pattern is signaled in advance to the device. Upon activation by L1/L2 control signaling, which also includes parameters such as the time—frequency resources and coding-and-modulation scheme to use, the device receives downlink data transmissions according to the preconfigured pattern.

In the downlink, semi-persistent scheduling is supported where the device is configured with a periodicity of the data transmissions using radio resource control (RRC) signaling. Activation of semi-persistent scheduling is done using the physical downlink control channel (PDCCH) as for dynamic scheduling but with the configured scheduling-radio network temporary identifier (CS-RNTI) instead of the normal cell-radio network temporary identifier (C-RNTI). The PDCCH also carries the necessary information in terms of time-frequency resources and other parameters needed in a similar way as dynamic scheduling. The hybrid-ARQ (HARQ) process number is derived from the time when the downlink data transmission starts according to a formula. Upon activation of semi-persistent scheduling, the device receives downlink data transmission periodically according to the RRC-configured periodicity using the transmission parameters indicated on the PDCCH activating the transmission. Hence, control signaling is only used once and the overhead is reduced. After enabling semi-persistent scheduling, the device continues to monitor the set of candidates PDCCHs for uplink and downlink scheduling commands. This is useful in the case that there are occasional transmissions of large amounts of data for which the semi-persistent allocation is not sufficient. It is also used to handle hybrid-ARQ retransmissions which are dynamically scheduled.

HARQ-ACK for (DL-SPS) physical downlink shared control channel (PDSCH)

There are two types of HARQ-ACK codebook construction:

1) Type 1: Semi-static codebook. The semi-static codebook can be viewed as a matrix consisting of a time-domain dimension and a component-carrier (or CBG or MIMO layer) dimension, both of which are semi-statically configured. In Semi-static the codebook size is fixed, the number

of bits to transmit in a hybrid-ARQ report is known and the appropriate format for the uplink control signaling can be selected. Not all transmission opportunities possible with the codebook are used in this example and for entries in the matrix without a corresponding transmission, a negative acknowledgment is transmitted. This provides robustness; in the case of missed downlink assignment a negative acknowledgment is provided to the gNB, which can retransmit the missing transport block (or CBG). One drawback with the semi-static codebook is the potentially large size of a hybrid-ARQ report. For a small number of component carriers and no CBG retransmissions, this is less of a problem, but if a large number of carriers and codeblock groups are configured out of which only a small number is simultaneously used, this may become more of an issue.

2) Type 2: Dynamic codebook. To address the drawback of a potentially large semi-static codebook size in some scenarios, NR also supports a dynamic codebook. In fact, this is the default codebook used unless the system is configured otherwise. With a dynamic codebook, only the acknowledgment information for the scheduled carriers is included in the report, instead of all carriers, scheduled or not, as is the case with a semi-static codebook. Hence, the size of the codebook is dynamically varying as a function of the number of scheduled carriers

Skipping Downlink Transmission in DL-SPS:

In Rel-16, DL-SPS can be configured with small periodicity down to one-slot periodicity. Due to the small configured periodicity, gNB may not have data in its buffer in some DL transmission occasions. In that case, gNB skips transmission to reduce DL interference on other UEs. This is shown in FIG. 2.

When gNB skips transmission in some transmission occasions (TO), the UE does not know whether in a TO data is transmitted or not and then the UE tries to report HARQ-ACK/NACK even for the non-transmitted TOs

One-Shot Feedback

The dynamic HARQ codebook in Rel-15 was used as a starting point for the design of an enhanced mechanism for operation on unlicensed spectrum. In Rel-15, the timing control for the PUCCH/UCI feedback is quite limited. For instance, if the UE fails to provide the aggregated feedback on the specified resources due to failed decoding or due to unsuccessful LBT specifically on unlicensed spectrum, all PDSCHs included in this feedback needs to be retransmitted. gNB does not have the flexibility to ask for feedback retransmission. The gNB will have to assume NACK and retransmit all the corresponding PDSCHs. Another issue is the PUCCH mis-detection at the gNB side. Even if the UE successfully transmits the HARQ feedback, there are chances that the gNB may not be able to detect it. From gNB perspective, failed LBT or missed UCI transmissions are indistinguishable. Due to the one-to-one mapping between PDSCH and corresponding feedback in the time domain, if the gNB fails to detect the feedback in the predefined time location, the gNB will have to assume NACK and retransmit all the corresponding PDSCHs. While missing PUCCH transmission on licensed carrier is unlikely, it is highly likely to happen on an unlicensed band due to collisions.

At least for NR-U operation, to solve the above issues, the UE can be configured to monitor feedback request of a HARQ-ACK codebook containing all DL HARQ processes. The feedback can be requested in DL DCI 1_1. In response to the trigger, the UE reports the HARQ ACK feedback for all DL HARQ processes. The format of the feedback,

3

CBG-based HARQ-ACK or TB-based HARQ-ACK, can be configured to be part of the one-shot HARQ feedback for the CCs configured with CBG.

Additionally, to resolve any possible ambiguity between the gNB and the UE that might be caused by possible mis-detection of PDCCH(s), the UE can be configured to reports the corresponding latest NDI value for a latest received PDSCH for that HARQ process along with the corresponding HARQ-ACK for the received PDSCH. from gNB perspective, If the NDI value matches the last transmitted value, it indicates that the reported HARQ-ACK feedback correctly corresponds to the HARQ process with pending feedback. Otherwise, the mismatch suggests that the UE is reporting an outdated feedback.

HARQ-ACK for DL-SPS in Rel-16

The following agreements have been done for Rel-16:

R1-1911602

Agreements:

For cases where only HARQ-ACK feedback for SPS PDSCHs without associated DL assignment shall be reported (i.e. no dynamic PDSCH HARQ-ACK),

Multiple PUCCH resources are configured common for all SPS configurations per HARQ-ACK codebook. The actual PUCCH resource to be used among PUCCH resources is determined based on HARQ-ACK payload size

Number of PUCCH resources is up to 4

FFS details (e.g., threshold for determining PUCCH resource)

R1-1911692

Agreements:

For cases where only HARQ-ACK feedback for SPS PDSCHs without associated DL assignment shall be reported (i.e. no dynamic PDSCH HARQ-ACK), PUCCH resource i is selected if HARQ-ACK payload size (not including CRC) is in the range of $\{N_{i,min}, \dots, N_{i,max}\}$ bits, where the number of PUCCH resources in the selection is from 0 up to 3.

$N_{0,min}=1, N_{0,max}=2$

For $i \neq 0$

$N_{i,max}$ is configured by RRC; if not configured, $N_{i,max}$ is 1706.

$N_{i,min}$ is equal to $N_{i-1,max}+1$

Note: The above mechanism is equivalent to rel-15 procedure when a single PUCCH resource is configured per PUCCH resource set.

R1-1912403 Summary of 7.2.6.7 Others LG Electronics

Agreements:

In Rel-16, multiple DL SPS configurations can be configured on different serving cells in a cell group.

Agreements:

If the UE is configured with more than one SPS PDSCH configurations, and for type-1 HARQ-ACK codebook construction,

For cases where HARQ-ACK feedback for one or more SPS PDSCH receptions without a corresponding PDCCH is multiplexed with HARQ-ACK feedback for dynamic scheduled PDSCH and/or for SPS PDSCH release, or

For cases where HARQ-ACK feedback for SPS PDSCH release is multiplexed with HARQ-ACK feedback for dynamic scheduled PDSCH, or

For cases where only HARQ-ACK feedback for SPS PDSCH release shall be reported,

4

HARQ-ACK bit location for SPS PDSCH reception is derived by reusing Rel-15 mechanism (i.e., based on the TDRA table row index and K1 indicated in the activation DCI)

HARQ-ACK bit location for SPS PDSCH release with a separate release DCI is derived by reusing Rel-15 mechanism (i.e., based on the TDRA table row index indicated in the activation DCI and K1 indicated in the release DCI)

HARQ-ACK bit location for SPS PDSCH release with a joint release DCI is derived based on the TDRA table row index indicated in the activation DCI for SPS PDSCH with the lowest SPS configuration index among the jointly released configurations and K1 indicated in the release DCI

Note: There is no change on the number of HARQ-ACK bits for a PUCCH transmission regardless of whether a joint release DCI is present or not.

In Rel-16, when the SPS configurations are released by a joint release DCI,

Multiple SPS configurations to be released by the joint release DCI should have the same priority

R1-1913451 Summary #2 of 7.2.6.7 Others LG Electronics

Agreement

For a rel-16 UE provided by SPS-PUCCH-AN-List a set of PUCCH resources, in case of collision between HARQ-ACK for SPS PDSCH without a corresponding PDCCH and SR for the same priority, reuse Rel-15 rule for collision between HARQ-ACK for dynamic scheduled PDSCH and SR in order to determine the PUCCH resource

Agreement

For a given SPS configuration activated by DCI format 1_2, the MCS table is determined by reusing Rel-15 mechanism for a SPS configuration activated by DCI format 1_1.

No new RRC parameter for mcs-Table is introduced for DCI format 1_2

Agreement

If the UE is configured with more than one SPS PDSCH configurations, for cases where only HARQ-ACK feedback for one or more SPS PDSCH receptions without a corresponding PDCCH shall be reported (i.e. no HARQ-ACK feedback for dynamic scheduled PDSCH and/or for SPS PDSCH release

HARQ-ACK bit order for SPS PDSCH reception without a corresponding PDCCH is determined

In ascending order of DL slot per {SPS configuration index, serving cell index}, and then in ascending order of SPS configuration index per {serving cell index}, and then in ascending order of serving cell index

Agreement

If the UE is configured with more than one SPS PDSCH configurations, and for type-2 HARQ-ACK codebook construction,

HARQ-ACK bit order for SPS PDSCH release with a separate/joint release DCI is derived by reusing rel-15 mechanism (i.e., based on DAI and K1 indicated in the release DCI)

HARQ-ACK bit order for SPS PDSCH with associated PDCCH is derived by reusing rel-15 mechanism (i.e., based on DAI and K1 indicated in the activation DCI)

For cases where HARQ-ACK feedback for one or more SPS PDSCH receptions without a corresponding PDCCH is multiplexed with HARQ-ACK feedback for dynamic scheduled PDSCH and/or for SPS PDSCH release,

5

HARQ-ACK for one or more SPS PDSCH receptions without a corresponding PDCCH is appended after HARQ-ACK bits for dynamic scheduled PDSCHs and/or for SPS PDSCH release

In ascending order of DL slot per {SPS configuration index, serving cell index}, and then in ascending order of SPS configuration index per {serving cell index}, and then in ascending order of serving cell index

Agreements: RAN1 #e100

For a UE not indicating a capability to receive more than one unicast PDSCH per slot, in a slot with more than one SPS PDSCHs each without a corresponding PDCCH and no dynamic scheduled PDSCH and/or for SPS PDSCH release, a UE is not required to receive SPS PDSCHs other than the SPS PDSCH with the lowest SPS configuration index among SPS PDSCHs in a slot (regardless of whether SPS PDSCHs are overlapped or not).

The UE shall report HARQ-ACK feedback only for the SPS PDSCH with the lowest SPS configuration index among SPS PDSCHs in the slot.

SUMMARY

When a UE is transmitting HARQ-ACK codebooks of fixed sizes, there are scenarios that the UE transmits the codebooks that would be unnecessary.

For example, in case of semi-static HARQ codebook, if the corresponding PDSCHs are not received properly, it is unnecessary to transmit NACK, since the gNB would assume the same in case of absence of the feedback.

Another example is for the case of DL SPS. In the current spec, a UE report HARQ ACK-NACK even if there is no data transmission in one DL-SPS transmission occasion.

Thus, the UE sends unnecessary HARQ-ACK/NACK feedback. This causes (a) wastage of UL resources, (b) causes interference to others' uplink transmissions and (c), causes wastage of UE's energy.

Besides, this mode of operation, where periodic transmissions are sent, is doomed inefficient in unlicensed spectrum where access to the channel is not always guaranteed.

Various embodiments of inventive concepts described herein skips (or reduce) HARQ-ACK feedback transmission for applicable to HARQ-ACK codebooks with fixed sizes such as HARQ-ACK feedback for DL SPS or semi-static (Type 1) HARQ codebook for scheduled transmission.

According to some embodiments of inventive concepts, a method for reporting HARQ-ACK code books of fixed sizes in a communication network in a user equipment, UE, communication device is provided. The method includes responsive to operating in a first mode, reporting the HARQ-ACK codebook for each scheduled transmission to a network node. The method further includes responsive to operating in a second mode, reporting the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is ACK, wherein no NACK is transmitted in the codebook when operating in the second mode. The method further includes responsive to operating in a third mode, reporting the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is NACK, wherein no ACK is transmitted in the codebook when operating in the second mode.

An advantage that is provided is that UL interference is reduced. Energy of the UE will be saved. Congestion on UL resource, e.g., PUCCH will be reduced which is beneficial

6

for operation on both licensed and unlicensed spectrum. Further a SPS PDSCH can be released for DL-SPS configured by one-slot periodicity.

Analogous communication devices, computer program, computer program products are also provided.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this application, illustrate certain non-limiting embodiments of inventive concepts. In the drawings:

FIG. 1 is an illustration of DL-SPS PDSCH transmission;

FIG. 2 is an illustration of skipping DL transmission in DL-SPS;

FIG. 3 is a block diagram illustrating a wireless device UE according to some embodiments of inventive concepts;

FIG. 4 is a block diagram illustrating a radio access network RAN node (e.g., a base station eNB/gNB) according to some embodiments of inventive concepts;

FIG. 5 is a block diagram illustrating a core network CN node (e.g., an AMF node, an SMF node, etc.) according to some embodiments of inventive concepts;

FIGS. 6-8 are flow charts illustrating operations of the wireless device UE according to some embodiments of inventive concepts;

FIG. 9 is a block diagram of a wireless network in accordance with some embodiments;

FIG. 10 is a block diagram of a user equipment in accordance with some embodiments

FIG. 11 is a block diagram of a virtualization environment in accordance with some embodiments;

FIG. 12 is a block diagram of a telecommunication network connected via an intermediate network to a host computer in accordance with some embodiments;

FIG. 13 is a block diagram of a host computer communicating via a base station with a user equipment over a partially wireless connection in accordance with some embodiments;

FIG. 14 is a block diagram of methods implemented in a communication system including a host computer, a base station and a user equipment in accordance with some embodiments;

FIG. 15 is a block diagram of methods implemented in a communication system including a host computer, a base station and a user equipment in accordance with some embodiments;

FIG. 16 is a block diagram of methods implemented in a communication system including a host computer, a base station and a user equipment in accordance with some embodiments; and

FIG. 17 is a block diagram of methods implemented in a communication system including a host computer, a base station and a user equipment in accordance with some embodiments.

DETAILED DESCRIPTION

The following description presents various embodiments of the disclosed subject matter. These embodiments are presented as teaching examples and are not to be construed as limiting the scope of the disclosed subject matter. For example, certain details of the described embodiments may be modified, omitted, or expanded upon without departing from the scope of the described subject matter.

FIG. 3 is a block diagram illustrating elements of a communication device UE 300 (also referred to as a mobile terminal, a mobile communication terminal, a wireless device, a wireless communication device, a wireless terminal, mobile device, a wireless communication terminal, user equipment, UE, a user equipment node/terminal/device, etc.) configured to provide wireless communication according to embodiments of inventive concepts. (Communication device 300 may be provided, for example, as discussed below with respect to wireless device 4110 of FIG. 9.) As shown, communication device UE may include an antenna 307 (e.g., corresponding to antenna 4111 of FIG. 9), and transceiver circuitry 301 (also referred to as a transceiver, e.g., corresponding to interface 4114 of FIG. 9) including a transmitter and a receiver configured to provide uplink and downlink radio communications with a base station(s) (e.g., corresponding to network node 4160 of FIG. 9, also referred to as a RAN node) of a radio access network. Communication device UE may also include processing circuitry 303 (also referred to as a processor, e.g., corresponding to processing circuitry 4120 of FIG. 9) coupled to the transceiver circuitry, and memory circuitry 305 (also referred to as memory, e.g., corresponding to device readable medium 4130 of FIG. 9) coupled to the processing circuitry. The memory circuitry 305 may include computer readable program code that when executed by the processing circuitry 303 causes the processing circuitry to perform operations according to embodiments disclosed herein. According to other embodiments, processing circuitry 303 may be defined to include memory so that separate memory circuitry is not required. Communication device UE may also include an interface (such as a user interface) coupled with processing circuitry 303, and/or communication device UE may be incorporated in a vehicle.

As discussed herein, operations of communication device UE may be performed by processing circuitry 303 and/or transceiver circuitry 301. For example, processing circuitry 303 may control transceiver circuitry 301 to transmit communications through transceiver circuitry 301 over a radio interface to a radio access network node (also referred to as a base station) and/or to receive communications through transceiver circuitry 301 from a RAN node over a radio interface. Moreover, modules may be stored in memory circuitry 305, and these modules may provide instructions so that when instructions of a module are executed by processing circuitry 303, processing circuitry 303 performs respective operations (e.g., operations discussed below with respect to Example Embodiments relating to wireless communication devices).

FIG. 4 is a block diagram illustrating elements of a radio access network RAN node 400 (also referred to as a network node, base station, eNodeB/eNB, gNodeB/gNB, etc.) of a Radio Access Network (RAN) configured to provide cellular communication according to embodiments of inventive concepts. (RAN node 400 may be provided, for example, as discussed below with respect to network node 4160 of FIG. 9.) As shown, the RAN node may include transceiver circuitry 401 (also referred to as a transceiver, e.g., corresponding to portions of interface 4190 of FIG. 9) including a transmitter and a receiver configured to provide uplink and downlink radio communications with mobile terminals. The RAN node may include network interface circuitry 407 (also referred to as a network interface, e.g., corresponding to portions of interface 4190 of FIG. 9) configured to provide communications with other nodes (e.g., with other base stations) of the RAN and/or core network CN. The network node may also include processing circuitry 403 (also

referred to as a processor, e.g., corresponding to processing circuitry 4170) coupled to the transceiver circuitry, and memory circuitry 405 (also referred to as memory, e.g., corresponding to device readable medium 4180 of FIG. 9) coupled to the processing circuitry. The memory circuitry 405 may include computer readable program code that when executed by the processing circuitry 403 causes the processing circuitry to perform operations according to embodiments disclosed herein. According to other embodiments, processing circuitry 403 may be defined to include memory so that a separate memory circuitry is not required.

As discussed herein, operations of the RAN node may be performed by processing circuitry 403, network interface 407, and/or transceiver 401. For example, processing circuitry 403 may control transceiver 401 to transmit downlink communications through transceiver 401 over a radio interface to one or more mobile terminals UEs and/or to receive uplink communications through transceiver 401 from one or more mobile terminals UEs over a radio interface. Similarly, processing circuitry 403 may control network interface 407 to transmit communications through network interface 407 to one or more other network nodes and/or to receive communications through network interface 407 from one or more other network nodes. Moreover, modules may be stored in memory 405, and these modules may provide instructions so that when instructions of a module are executed by processing circuitry 403, processing circuitry 403 performs respective operations (e.g., operations discussed below with respect to Example Embodiments relating to RAN nodes).

According to some other embodiments, a network node may be implemented as a core network CN node without a transceiver. In such embodiments, transmission to a wireless communication device UE may be initiated by the network node so that transmission to the wireless communication device UE is provided through a network node including a transceiver (e.g., through a base station or RAN node). According to embodiments where the network node is a RAN node including a transceiver, initiating transmission may include transmitting through the transceiver.

FIG. 5 is a block diagram illustrating elements of a core network CN node (e.g., an SMF node, an AMF node, etc.) of a communication network configured to provide cellular communication according to embodiments of inventive concepts. As shown, the CN node may include network interface circuitry 507 (also referred to as a network interface) configured to provide communications with other nodes of the core network and/or the radio access network RAN. The CN node may also include a processing circuitry 503 (also referred to as a processor) coupled to the network interface circuitry, and memory circuitry 505 (also referred to as memory) coupled to the processing circuitry. The memory circuitry 505 may include computer readable program code that when executed by the processing circuitry 503 causes the processing circuitry to perform operations according to embodiments disclosed herein. According to other embodiments, processing circuitry 503 may be defined to include memory so that a separate memory circuitry is not required.

As discussed herein, operations of the CN node may be performed by processing circuitry 503 and/or network interface circuitry 507. For example, processing circuitry 503 may control network interface circuitry 507 to transmit communications through network interface circuitry 507 to one or more other network nodes and/or to receive communications through network interface circuitry 507 from one or more other network nodes. Moreover, modules may be stored in memory 505, and these modules may provide

instructions so that when instructions of a module are executed by processing circuitry 503, processing circuitry 503 performs respective operations (e.g., operations discussed below with respect to Example Embodiments relating to core network nodes).

Various embodiments of inventive concepts shall now be described.

Embodiment 1

In this embodiment, 3 modes of operation for reporting HARQ-ACK code books of fixed sizes are assumed.

1. Mode 1: No HARQ skipping (Legacy behaviour)

This is the legacy behaviour.

2. Mode 2: Skip All-NACK

In this mode, the UE reports the HARQ-ACK codebook if at least one of the HARQ-ACK information in the codebook is ACK.

3. Mode 3: Skip All-ACK

In this mode, the UE reports the HARQ-ACK codebook if at least one of the HARQ-ACK information in the codebook is NACK.

if the SPS PDSCH is not decoded/detected successfully, UE skips sending NACK in the PUCCH occasion corresponding to the SPS PDSCH reception. Otherwise, UE sends ACK.

The UE is indicated (via RRC configuration or dynamically via DCI) send HARQ-ACK feedback corresponding to one of the modes of operations described above.

In one example, the default mode of operation can be mode one. However, the UE can change to another mode, by means of RRC configuration or dynamic signaling.

In another example, the UE operates in one of the modes until a corresponding timer expires. The timer can be configured by higher layers.

In another example, operation in a mode is determined by the deployment scenario or base on the collected statistics on UE experience. For example, in scenario with very reliable links, Mode 3 is preferred and Mode 2, otherwise. The decision can be based on percentages of Feedback error rate.

In another example, operation in a mode, can be determined by a time pattern, or a frequency pattern (e.g. a set of cells or BWPs or combination of both). The patterns can be RRC configured or indicated dynamically to the UE based on pre-configuration, using fields in DCI. The DCI could be UE specific or common between groups of UEs.

In another example, the UE can be enabled by any of the modes above, when the modes are applicable to different time or frequency or spatial resources.

In case of DL SPS, the mode of operations can be part of a DL SPS configuration, or common between a group or multiple groups of configurations. In another example, a mode of operation can be enabled by an activation DCI. In another example, for DL SPS configurations contributing to the same HARQ codebook being reported in a PUCCH, if at least of the corresponding activation DCI indicate enabling a change of mode, that would be applicable to other configurations involved in the same HARQ codebook.

Embodiment 2: Using Type 3 One Shot Feedback

Type 3 HARQ codebook is used in combination with any of the modes listed in embodiment 1. Then gNB can request for feedback from UE for all configured HARQ processes.

As an extension, Type 3 codebook is extended to support triggering for a subset of HARQ processes instead of all HARQ processes. As a non-limiting example, the UE is

configured to report the feedback for SPS configured HARQ processes in Type 3 HARQ-codebook.

To determine the HARQ feedback to be reported, one or more of the following variances is considered:

The UE reports ACK/NACK corresponding to the latest obtained feedback for SPS PDSCH occasion for a certain HARQ process occurrence before the PUCCH occasion where Type 3 codebook should be sent.

If no PDSCH was detected in that occurrence, NACK is reported

If the UE previously sent the feedback to the gNB, NACK is reported.

The SPS PDSCH occurrences within a past time window is considered. The time window can be RRC configured.

The start or end of time window with respect to one-shot DCI and duration of the time window can be configured.

Embodiment 3

The UE is configured to send HARQ-feedback report that includes feedback corresponding to the SPS configured HARQ processes periodically.

As another variant, the UE is configured to send HARQ-feedback for multiple SPS PDSCH occasion in one PUCCH. The report is sent periodically and includes the feedback for all SPS PDSCH occasions since the last report.

For example, UE collects HARQ-bits for N SPS occasions and transmit in the HARQ-CB, and then collects HARQ-bits for next N SPS occasion and transmits in the corresponding HARQ-CB.

If PDSCH allocations are repeated in a periodic manner (measured in time-units), the periodicity of HARQ-CB is integer multiple of PDSCH's periodicity, i.e., if PDSCH occasion is repeated after every 3 slots, then to deliver feedback for N PDSCHs in one CB, the HARQ-CB is allocated after every 3N slots. Further, the HARQ-CB periodicity (i.e., N, can be configured via RRC or can be dynamically changed through DCI signalling. In addition, the periodic feedback reporting can be activated or deactivated via DCI signalling. In Embodiment 2, the one-shot feedback is trigger only by DCI, here one-feedback is configured in a periodic manner (not by DCI), but the parameters (e.g., K1, periodicity N) of periodic reporting can be configured via RRC or DCI or combination of RRC and DCI.

Embodiment 4

In one embodiment, if UE detects DMRS or not, and further if data is correctly decoded or not given DMRS is decoded over the SPS PDSCH allocation, then UE's feedback transmission depends on the selected mode from the following.

1. Only ACK-based feedback allowed, or
2. Only NACK-based feedback allowed, or
3. Both ACK and NACK based feedback allowed, or
4. No feedback allowed

For example, if mode 1 is selected then UE sends ACK for correctly decoding PDSCH (including for false alarms) or sends NACK otherwise (i.e., for cases with no detection or misdetection, or even if DMRS is detected but data is not).

If mode 2 is selected, then UE transmits ACK (like mode 1) but skips NACK (unlike mode 1).

If mode 3 is selected, then UE transmits NACK (like mode 1) but skips ACK (unlike mode 1)

11

If mode 4 is selected, then UE does not transmit any feedback.

Embodiment 5

The HARQ-feedback UCI (any type) can be sent in the PUCCH or in the PUSCH (applicable to all embodiment)

Embodiment 6

If PUCCH carrying HARQ feedback corresponding to an SPS PDSCH reception collides with PUCCH carrying Type 3 HARQ codebook, the UE sends only the Type3 HARQ codebook

Embodiment 7: In one embodiment, if the DCI has priority configured, then it will provide feedback for indicated priority SPSs' PDSCHs in the CB. In RRC a table can be configured where high DCI priority maps to certain SPS indices and low DCI priority maps to other SPS indices.

Embodiment 7a: In one embodiment, the high priority DCI asking one-shot based feedback can override low priority DCI.

Embodiment 8: In one embodiment, the DCI asking one-shot based feedback can be transmitted in PDSCHs.

Embodiment 9: In one embodiment, the HARQ-bits in the CB can be arranged with following options,

1. According to the HARQ processes, or
2. According to SPS indices and then according to HARQ processes per SPS index.

In the description that follows, while the communication device may be any of the communication device 300, wireless device 4110, the UE 4200, UEs 4491, 4492, virtual hardware 4330, virtual machine 4340, or UE 4530, the communication device 300 shall be used to describe the functionality of the operations of the communication device. Operations of the communication device 300 (implemented using the structure of the block diagram of FIG. 3) will now be discussed with reference to the flow chart of FIG. 6 according to some embodiments of inventive concepts. For example, modules may be stored in memory 305 of FIG. 3, and these modules may provide instructions so that when the instructions of a module are executed by respective communication device processing circuitry 303, processing circuitry 303 performs respective operations of the flow chart.

FIG. 6 illustrates embodiments of inventive concepts of reporting hybrid automatic repeat request-acknowledgement (HARQ-ACK) code books of fixed sizes in a communication network in a user equipment, UE, communication device 300.

Turning to FIG. 6, in block 601, the processing circuitry 303, responsive to operating in a first mode, reports the HARQ-ACK codebook for each scheduled transmission to a network node. The first mode is described above at least in embodiment 1.

In block 603, the processing circuitry 303, responsive to operating in a second mode, reports the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is ACK. The second mode is described above at least in embodiment 1.

Thus, as described above, no NACK is reported to the network node. In other words, the processing circuitry 303, responsive to operating in a second mode, reports the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is ACK, wherein no NACK is transmitted in the codebook when operating in the second mode.

12

In block 605, the processing circuitry 303, responsive to operating in a third mode, reports the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is negative acknowledgement (NACK).

Thus, as described above, no ACK is reported to the network node. In other words, the processing circuitry 303, responsive to operating in a third mode, reports the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is NACK, wherein no ACK is transmitted in the codebook when operating in the second mode.

In some embodiments of inventive concepts, responsive to a semi persistent scheduling (SPS) physical downlink shared channel (PDSCH) not be decoded/detected successfully, the processing circuitry 303 skips sending NACK in the physical uplink control channel, PUCCH, occasion corresponding to the SPS PDSCH reception.

In other embodiments of inventive concepts, the processing circuitry 303 configures the UE communication device to periodically send, to the network node, HARQ-feedback report that includes feedback corresponding to the SPS configured HARQ processes.

In various embodiments of inventive concepts, the processing circuitry 303 sends HARQ-feedback uplink control information, UCI, of any type to the network node in the PUCCH or in the PUSCH.

In further embodiments, the processing circuitry 303, responsive to a PUCCH carrying HARQ feedback corresponding to an SPS PDSCH reception colliding with a PUCCH carrying Type3 HARQ codebook, transmits only the Type3 HARQ codebook to the network node.

In other embodiments of inventive concepts, the processing circuitry 303, responsive to the downlink control information, DCI, having priority configured, provides feedback to the network node for high priority SPSs' PDSCHs in the codebook. For example, high priority DCI asking one-shot based feedback can override low priority DCI. The DCI asking one-shot based feedback can be transmitted in PDSCHs.

In yet other embodiments of inventive concepts, the processing circuitry 303 arranges HARQ-bits in the codebook according to the HARQ process or according to SPS indices and then according to HARQ processes per SPS index.

FIG. 7 illustrates embodiments of inventive concepts where a Type 3 HARQ codebook is used in combination with any of the first mode, the second mode, and the third mode.

Turning to FIG. 7, in block 701, the processing circuitry 303 extends the Type 3 codebook to support triggering for a subset of HARQ processes instead of all HARQ processes. This can reduce the number of transmissions being made.

In block 703, the processing circuitry 303 determines a HARQ feedback to be reported. In some embodiments, the processing circuitry 303, in block 705, reports ACK/NACK to the network node corresponding to a latest obtained feedback for SPS PDSCH occasion for a certain HARQ process occurrence before the PUCCH occasion where Type 3 codebook should be sent. In other embodiments, the processing circuitry 303 reports ACK/NACK based on a number of SPS PDSCH occurrences within a past time window.

FIG. 8 illustrates embodiments of inventive concepts where if the communicate device 300 detects a demodulation reference signal (DMRS) or not, and further if data is correctly decoded or not given DMRS is decoded over the

13

SPS PDSCH allocation, then UE's feedback transmission depends on the selected mode.

Turing to FIG. 8, in block 801, the processing circuitry 303, responsive to operating in the first mode, transmits, to the network node, a NACK for cases with no detection or misdetection, or if demodulation reference signal, DMRS, is detected but data is not.

In block 803, the processing circuitry 303, responsive to operating in the second mode, transmits, to the network node, the ACK responsive to correctly decoding PDSCH and not transmitting, to the network node, a NACK for cases with no detection or misdetection, or if DMRS is detected but data is not.

In block 805, the processing circuitry 303, responsive to operating in a third mode, does not transmit, to the network node, an ACK responsive to correctly decoding PDSCH and transmitting, to the network node, a NACK for cases with no detection or misdetection, or if DMRS is detected but data is not.

In block 807, the processing circuitry 303, responsive to operating in a fourth mode, transmits no feedback to the network node.

Example embodiments are discussed below.

1. A method for reporting HARQ-ACK code books of fixed sizes in a communication network in a user equipment, UE, communication device (300), the method comprising:

responsive to operating in a first mode, reporting (601) the HARQ-ACK codebook for each scheduled transmission to a network node;

responsive to operating in a second mode, reporting (603) the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is ACK;

responsive to operating in a third mode, reporting (605) the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is NACK.

2. The method of Embodiment 1, further comprising: responsive to a SPS PDSCH not be decoded/detected successfully, skips sending NACK in the PUCCH occasion corresponding to the SPS PDSCH reception.

3. The method of any of Embodiments 1-2 wherein a Type 3 HARQ codebook is used in combination with any of the first mode, the second mode, and the third mode, the method further comprising:

extending the Type 3 codebook to support triggering for a subset of HARQ processes instead of all HARQ processes.

4. The method of Embodiment 3, further comprising determining a HARQ feedback to be reported by at least one of:

reporting ACK/NACK to the network node corresponding to a latest obtained feedback for SPS PDSCH occasion for a certain HARQ process occurrence before the PUCCH occasion where Type 3 codebook should be sent; and

reporting ACK/NACK based on a number of SPS PDSCH occurrences within a past time window.

5. The method of any of Embodiments 1-2, further comprising:

configuring the UE communication device to periodically send, to the network node, HARQ-feedback report that includes feedback corresponding to the SPS configured HARQ processes.

14

6. The method of any of Embodiments 1-2, further comprising:

responsive to operating in the first mode, transmitting (801), to the network node, an ACK responsive to correctly decoding PDSCH and transmitting, to the network node, a NACK for cases with no detection or misdetection, or if DMRS is detected but data is not;

responsive to operating in the second mode, transmitting (803), to the network node, the ACK responsive to correctly decoding PDSCH and not transmitting, to the network node, a NACK for cases with no detection or misdetection, or if DMRS is detected but data is not;

responsive to operating in a third mode, not transmitting (805), to the network node, an ACK responsive to correctly decoding PDSCH and transmitting, to the network node, a NACK for cases with no detection or misdetection, or if DMRS is detected but data is not; and

responsive to operating in a fourth mode, transmitting (807) no feedback to the network node.

7. The method of any of Embodiments 1-6, wherein the HARQ-feedback UCI of any type is sent to the network node in the PUCCH or in the PUSCH.

8. The method of any of Embodiments 1-2, further comprising:

responsive to a PUCCH carrying HARQ feedback corresponding to an SPS PDSCH reception colliding with a PUCCH carrying Type3 HARQ codebook, transmitting only the Type3 HARQ codebook to the network node.

9. The method of any of Embodiments 1-2, wherein responsive to the DCI having priority configured, providing feedback to the network node for high priority SPSs' PDSCHs in the codebook.

10. The method of any of Embodiments 1-9, wherein high priority DCI asking one-shot based feedback can override low priority DCI.

11. The method of Embodiment 10 wherein the DCI asking one-shot based feedback is transmitted in PDSCHs

12. The method of any of Embodiments 1-11, further comprising arranging HARQ-bits in the codebook according to the HARQ process or according to SPS indices and then according to HARQ processes per SPS index.

13. A communication device (300) comprising:

processing circuitry (303); and

memory (305) coupled with the processing circuitry, wherein the memory includes instructions that when executed by the processing circuitry causes the communication device to perform operations according to any of Embodiments 1-13.

14. A communication device (300) adapted to perform according to any of Embodiments 1-13.

15. A computer program comprising program code to be executed by processing circuitry (303) of a communication device (300), whereby execution of the program code causes the communication device (300) to perform operations according to any of Embodiments 1-13.

16. A computer program product comprising a non-transitory storage medium including program code to be executed by processing circuitry (303) of a communication device (300), whereby execution of the program code causes the communication device (300) to perform operations according to any of embodiments 1-13.

Explanations are provided below for various abbreviations/acronyms used in the present disclosure.

Abbreviation	Explanation
ACK	Acknowledgement
ARQ	Automatic Repeat Request
DMRS	Demodulation Reference Signal
NACK	Negative Acknowledgement
HARQ	Hybrid ARQ

References are identified below.

Additional explanation is provided below.

Generally, all terms used herein are to be interpreted according to their ordinary meaning in the relevant technical field, unless a different meaning is clearly given and/or is implied from the context in which it is used. All references to a/an/the element, apparatus, component, means, step, etc. are to be interpreted openly as referring to at least one instance of the element, apparatus, component, means, step, etc., unless explicitly stated otherwise. The steps of any methods disclosed herein do not have to be performed in the exact order disclosed, unless a step is explicitly described as following or preceding another step and/or where it is implicit that a step must follow or precede another step. Any feature of any of the embodiments disclosed herein may be applied to any other embodiment, wherever appropriate. Likewise, any advantage of any of the embodiments may apply to any other embodiments, and vice versa. Other objectives, features and advantages of the enclosed embodiments will be apparent from the following description.

Some of the embodiments contemplated herein will now be described more fully with reference to the accompanying drawings. Other embodiments, however, are contained within the scope of the subject matter disclosed herein, the disclosed subject matter should not be construed as limited to only the embodiments set forth herein; rather, these embodiments are provided by way of example to convey the scope of the subject matter to those skilled in the art.

FIG. 9 illustrates a wireless network in accordance with some embodiments.

Although the subject matter described herein may be implemented in any appropriate type of system using any suitable components, the embodiments disclosed herein are described in relation to a wireless network, such as the example wireless network illustrated in FIG. 9. For simplicity, the wireless network of FIG. 9 only depicts network **4106**, network nodes **4160** and **4160b**, and WDs **4110**, **4110b**, and **4110c** (also referred to as mobile terminals). In practice, a wireless network may further include any additional elements suitable to support communication between wireless devices or between a wireless device and another communication device, such as a landline telephone, a service provider, or any other network node or end device. Of the illustrated components, network node **4160** and wireless device (WD) **4110** are depicted with additional detail. The wireless network may provide communication and other types of services to one or more wireless devices to facilitate the wireless devices' access to and/or use of the services provided by, or via, the wireless network.

The wireless network may comprise and/or interface with any type of communication, telecommunication, data, cellular, and/or radio network or other similar type of system. In some embodiments, the wireless network may be configured to operate according to specific standards or other types of predefined rules or procedures. Thus, particular embodiments of the wireless network may implement communication standards, such as Global System for Mobile Communications (GSM), Universal Mobile Telecommuni-

cations System (UMTS), Long Term Evolution (LTE), and/or other suitable 2G, 3G, 4G, or 5G standards; wireless local area network (WLAN) standards, such as the IEEE 802.11 standards; and/or any other appropriate wireless communication standard, such as the Worldwide Interoperability for Microwave Access (WiMax), Bluetooth, Z-Wave and/or ZigBee standards.

Network **4106** may comprise one or more backhaul networks, core networks, IP networks, public switched telephone networks (PSTNs), packet data networks, optical networks, wide-area networks (WANs), local area networks (LANs), wireless local area networks (WLANs), wired networks, wireless networks, metropolitan area networks, and other networks to enable communication between devices.

Network node **4160** and WD **4110** comprise various components described in more detail below. These components work together in order to provide network node and/or wireless device functionality, such as providing wireless connections in a wireless network. In different embodiments, the wireless network may comprise any number of wired or wireless networks, network nodes, base stations, controllers, wireless devices, relay stations, and/or any other components or systems that may facilitate or participate in the communication of data and/or signals whether via wired or wireless connections.

As used herein, network node refers to equipment capable, configured, arranged and/or operable to communicate directly or indirectly with a wireless device and/or with other network nodes or equipment in the wireless network to enable and/or provide wireless access to the wireless device and/or to perform other functions (e.g., administration) in the wireless network. Examples of network nodes include, but are not limited to, access points (APs) (e.g., radio access points), base stations (BSs) (e.g., radio base stations, Node Bs, evolved Node Bs (eNBs) and NR NodeBs (gNBs)). Base stations may be categorized based on the amount of coverage they provide (or, stated differently, their transmit power level) and may then also be referred to as femto base stations, pico base stations, micro base stations, or macro base stations. A base station may be a relay node or a relay donor node controlling a relay. A network node may also include one or more (or all) parts of a distributed radio base station such as centralized digital units and/or remote radio units (RRUs), sometimes referred to as Remote Radio Heads (RRHs). Such remote radio units may or may not be integrated with an antenna as an antenna integrated radio. Parts of a distributed radio base station may also be referred to as nodes in a distributed antenna system (DAS). Yet further examples of network nodes include multi-standard radio (MSR) equipment such as MSR BSs, network controllers such as radio network controllers (RNCs) or base station controllers (BSCs), base transceiver stations (BTSs), transmission points, transmission nodes, multi-cell/multicast coordination entities (MCEs), core network nodes (e.g., MSCs, MMEs), O&M nodes, OSS nodes, SON nodes, positioning nodes (e.g., E-SMLCs), and/or MDTs. As another example, a network node may be a virtual network node as described in more detail below. More generally, however, network nodes may represent any suitable device (or group of devices) capable, configured, arranged, and/or operable to enable and/or provide a wireless device with access to the wireless network or to provide some service to a wireless device that has accessed the wireless network.

In FIG. 9, network node **4160** includes processing circuitry **4170**, device readable medium **4180**, interface **4190**, auxiliary equipment **4184**, power source **4186**, power cir-

17

cuitry **4187**, and antenna **4162**. Although network node **4160** illustrated in the example wireless network of FIG. **9** may represent a device that includes the illustrated combination of hardware components, other embodiments may comprise network nodes with different combinations of components. It is to be understood that a network node comprises any suitable combination of hardware and/or software needed to perform the tasks, features, functions and methods disclosed herein. Moreover, while the components of network node **4160** are depicted as single boxes located within a larger box, or nested within multiple boxes, in practice, a network node may comprise multiple different physical components that make up a single illustrated component (e.g., device readable medium **4180** may comprise multiple separate hard drives as well as multiple RAM modules).

Similarly, network node **4160** may be composed of multiple physically separate components (e.g., a NodeB component and a RNC component, or a BTS component and a BSC component, etc.), which may each have their own respective components. In certain scenarios in which network node **4160** comprises multiple separate components (e.g., BTS and BSC components), one or more of the separate components may be shared among several network nodes. For example, a single RNC may control multiple NodeB's. In such a scenario, each unique NodeB and RNC pair, may in some instances be considered a single separate network node. In some embodiments, network node **4160** may be configured to support multiple radio access technologies (RATs). In such embodiments, some components may be duplicated (e.g., separate device readable medium **4180** for the different RATs) and some components may be reused (e.g., the same antenna **4162** may be shared by the RATs). Network node **4160** may also include multiple sets of the various illustrated components for different wireless technologies integrated into network node **4160**, such as, for example, GSM, WCDMA, LTE, NR, WiFi, or Bluetooth wireless technologies. These wireless technologies may be integrated into the same or different chip or set of chips and other components within network node **4160**.

Processing circuitry **4170** is configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being provided by a network node. These operations performed by processing circuitry **4170** may include processing information obtained by processing circuitry **4170** by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored in the network node, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination.

Processing circuitry **4170** may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuit, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software and/or encoded logic operable to provide, either alone or in conjunction with other network node **4160** components, such as device readable medium **4180**, network node **4160** functionality. For example, processing circuitry **4170** may execute instructions stored in device readable medium **4180** or in memory within processing circuitry **4170**. Such functionality may include providing any of the various wireless features, functions, or benefits discussed herein. In some embodiments, processing circuitry **4170** may include a system on a chip (SOC).

18

In some embodiments, processing circuitry **4170** may include one or more of radio frequency (RF) transceiver circuitry **4172** and baseband processing circuitry **4174**. In some embodiments, radio frequency (RF) transceiver circuitry **4172** and baseband processing circuitry **4174** may be on separate chips (or sets of chips), boards, or units, such as radio units and digital units. In alternative embodiments, part or all of RF transceiver circuitry **4172** and baseband processing circuitry **4174** may be on the same chip or set of chips, boards, or units.

In certain embodiments, some or all of the functionality described herein as being provided by a network node, base station, eNB or other such network device may be performed by processing circuitry **4170** executing instructions stored on device readable medium **4180** or memory within processing circuitry **4170**. In alternative embodiments, some or all of the functionality may be provided by processing circuitry **4170** without executing instructions stored on a separate or discrete device readable medium, such as in a hard-wired manner. In any of those embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry **4170** can be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry **4170** alone or to other components of network node **4160**, but are enjoyed by network node **4160** as a whole, and/or by end users and the wireless network generally.

Device readable medium **4180** may comprise any form of volatile or non-volatile computer readable memory including, without limitation, persistent storage, solid-state memory, remotely mounted memory, magnetic media, optical media, random access memory (RAM), read-only memory (ROM), mass storage media (for example, a hard disk), removable storage media (for example, a flash drive, a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device readable and/or computer-executable memory devices that store information, data, and/or instructions that may be used by processing circuitry **4170**. Device readable medium **4180** may store any suitable instructions, data or information, including a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry **4170** and, utilized by network node **4160**. Device readable medium **4180** may be used to store any calculations made by processing circuitry **4170** and/or any data received via interface **4190**. In some embodiments, processing circuitry **4170** and device readable medium **4180** may be considered to be integrated.

Interface **4190** is used in the wired or wireless communication of signalling and/or data between network node **4160**, network **4106**, and/or WDs **4110**. As illustrated, interface **4190** comprises port(s)/terminal(s) **4194** to send and receive data, for example to and from network **4106** over a wired connection. Interface **4190** also includes radio front end circuitry **4192** that may be coupled to, or in certain embodiments a part of, antenna **4162**. Radio front end circuitry **4192** comprises filters **4198** and amplifiers **4196**. Radio front end circuitry **4192** may be connected to antenna **4162** and processing circuitry **4170**. Radio front end circuitry may be configured to condition signals communicated between antenna **4162** and processing circuitry **4170**. Radio front end circuitry **4192** may receive digital data that is to be sent out to other network nodes or WDs via a wireless connection. Radio front end circuitry **4192** may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters

4198 and/or amplifiers **4196**. The radio signal may then be transmitted via antenna **4162**. Similarly, when receiving data, antenna **4162** may collect radio signals which are then converted into digital data by radio front end circuitry **4192**. The digital data may be passed to processing circuitry **4170**. In other embodiments, the interface may comprise different components and/or different combinations of components.

In certain alternative embodiments, network node **4160** may not include separate radio front end circuitry **4192**, instead, processing circuitry **4170** may comprise radio front end circuitry and may be connected to antenna **4162** without separate radio front end circuitry **4192**. Similarly, in some embodiments, all or some of RF transceiver circuitry **4172** may be considered a part of interface **4190**. In still other embodiments, interface **4190** may include one or more ports or terminals **4194**, radio front end circuitry **4192**, and RF transceiver circuitry **4172**, as part of a radio unit (not shown), and interface **4190** may communicate with base-band processing circuitry **4174**, which is part of a digital unit (not shown).

Antenna **4162** may include one or more antennas, or antenna arrays, configured to send and/or receive wireless signals. Antenna **4162** may be coupled to radio front end circuitry **4192** and may be any type of antenna capable of transmitting and receiving data and/or signals wirelessly. In some embodiments, antenna **4162** may comprise one or more omni-directional, sector or panel antennas operable to transmit/receive radio signals between, for example, 2 GHz and 66 GHz. An omni-directional antenna may be used to transmit/receive radio signals in any direction, a sector antenna may be used to transmit/receive radio signals from devices within a particular area, and a panel antenna may be a line of sight antenna used to transmit/receive radio signals in a relatively straight line. In some instances, the use of more than one antenna may be referred to as MIMO. In certain embodiments, antenna **4162** may be separate from network node **4160** and may be connectable to network node **4160** through an interface or port.

Antenna **4162**, interface **4190**, and/or processing circuitry **4170** may be configured to perform any receiving operations and/or certain obtaining operations described herein as being performed by a network node. Any information, data and/or signals may be received from a wireless device, another network node and/or any other network equipment. Similarly, antenna **4162**, interface **4190**, and/or processing circuitry **4170** may be configured to perform any transmitting operations described herein as being performed by a network node. Any information, data and/or signals may be transmitted to a wireless device, another network node and/or any other network equipment.

Power circuitry **4187** may comprise, or be coupled to, power management circuitry and is configured to supply the components of network node **4160** with power for performing the functionality described herein. Power circuitry **4187** may receive power from power source **4186**. Power source **4186** and/or power circuitry **4187** may be configured to provide power to the various components of network node **4160** in a form suitable for the respective components (e.g., at a voltage and current level needed for each respective component). Power source **4186** may either be included in, or external to, power circuitry **4187** and/or network node **4160**. For example, network node **4160** may be connectable to an external power source (e.g., an electricity outlet) via an input circuitry or interface such as an electrical cable, whereby the external power source supplies power to power circuitry **4187**. As a further example, power source **4186** may comprise a source of power in the form of a battery or

battery pack which is connected to, or integrated in, power circuitry **4187**. The battery may provide backup power should the external power source fail. Other types of power sources, such as photovoltaic devices, may also be used.

Alternative embodiments of network node **4160** may include additional components beyond those shown in FIG. **9** that may be responsible for providing certain aspects of the network node's functionality, including any of the functionality described herein and/or any functionality necessary to support the subject matter described herein. For example, network node **4160** may include user interface equipment to allow input of information into network node **4160** and to allow output of information from network node **4160**. This may allow a user to perform diagnostic, maintenance, repair, and other administrative functions for network node **4160**.

As used herein, wireless device (WD) refers to a device capable, configured, arranged and/or operable to communicate wirelessly with network nodes and/or other wireless devices. Unless otherwise noted, the term WD may be used interchangeably herein with user equipment (UE). Communicating wirelessly may involve transmitting and/or receiving wireless signals using electromagnetic waves, radio waves, infrared waves, and/or other types of signals suitable for conveying information through air. In some embodiments, a WD may be configured to transmit and/or receive information without direct human interaction. For instance, a WD may be designed to transmit information to a network on a predetermined schedule, when triggered by an internal or external event, or in response to requests from the network. Examples of a WD include, but are not limited to, a smart phone, a mobile phone, a cell phone, a voice over IP (VoIP) phone, a wireless local loop phone, a desktop computer, a personal digital assistant (PDA), a wireless cameras, a gaming console or device, a music storage device, a playback appliance, a wearable terminal device, a wireless endpoint, a mobile station, a tablet, a laptop, a laptop-embedded equipment (LEE), a laptop-mounted equipment (LME), a smart device, a wireless customer-premise equipment (CPE), a vehicle-mounted wireless terminal device, etc. A WD may support device-to-device (D2D) communication, for example by implementing a 3GPP standard for sidelink communication, vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I), vehicle-to-everything (V2X) and may in this case be referred to as a D2D communication device. As yet another specific example, in an Internet of Things (IoT) scenario, a WD may represent a machine or other device that performs monitoring and/or measurements, and transmits the results of such monitoring and/or measurements to another WD and/or a network node. The WD may in this case be a machine-to-machine (M2M) device, which may in a 3GPP context be referred to as an MTC device. As one particular example, the WD may be a UE implementing the 3GPP narrow band internet of things (NB-IoT) standard. Particular examples of such machines or devices are sensors, metering devices such as power meters, industrial machinery, or home or personal appliances (e.g. refrigerators, televisions, etc.) personal wearables (e.g., watches, fitness trackers, etc.). In other scenarios, a WD may represent a vehicle or other equipment that is capable of monitoring and/or reporting on its operational status or other functions associated with its operation. A WD as described above may represent the endpoint of a wireless connection, in which case the device may be referred to as a wireless terminal. Furthermore, a WD as described above may be mobile, in which case it may also be referred to as a mobile device or a mobile terminal.

21

As illustrated, wireless device **4110** includes antenna **4111**, interface **4114**, processing circuitry **4120**, device readable medium **4130**, user interface equipment **4132**, auxiliary equipment **4134**, power source **4136** and power circuitry **4137**. WD **4110** may include multiple sets of one or more of the illustrated components for different wireless technologies supported by WD **4110**, such as, for example, GSM, WCDMA, LTE, NR, WiFi, WiMAX, or Bluetooth wireless technologies, just to mention a few. These wireless technologies may be integrated into the same or different chips or set of chips as other components within WD **4110**.

Antenna **4111** may include one or more antennas or antenna arrays, configured to send and/or receive wireless signals, and is connected to interface **4114**. In certain alternative embodiments, antenna **4111** may be separate from WD **4110** and be connectable to WD **4110** through an interface or port. Antenna **4111**, interface **4114**, and/or processing circuitry **4120** may be configured to perform any receiving or transmitting operations described herein as being performed by a WD. Any information, data and/or signals may be received from a network node and/or another WD. In some embodiments, radio front end circuitry and/or antenna **4111** may be considered an interface.

As illustrated, interface **4114** comprises radio front end circuitry **4112** and antenna **4111**. Radio front end circuitry **4112** comprise one or more filters **4118** and amplifiers **4116**. Radio front end circuitry **4112** is connected to antenna **4111** and processing circuitry **4120**, and is configured to condition signals communicated between antenna **4111** and processing circuitry **4120**. Radio front end circuitry **4112** may be coupled to or a part of antenna **4111**. In some embodiments, WD **4110** may not include separate radio front end circuitry **4112**; rather, processing circuitry **4120** may comprise radio front end circuitry and may be connected to antenna **4111**. Similarly, in some embodiments, some or all of RF transceiver circuitry **4122** may be considered a part of interface **4114**. Radio front end circuitry **4112** may receive digital data that is to be sent out to other network nodes or WDs via a wireless connection. Radio front end circuitry **4112** may convert the digital data into a radio signal having the appropriate channel and bandwidth parameters using a combination of filters **4118** and/or amplifiers **4116**. The radio signal may then be transmitted via antenna **4111**. Similarly, when receiving data, antenna **4111** may collect radio signals which are then converted into digital data by radio front end circuitry **4112**. The digital data may be passed to processing circuitry **4120**. In other embodiments, the interface may comprise different components and/or different combinations of components.

Processing circuitry **4120** may comprise a combination of one or more of a microprocessor, controller, microcontroller, central processing unit, digital signal processor, application-specific integrated circuit, field programmable gate array, or any other suitable computing device, resource, or combination of hardware, software, and/or encoded logic operable to provide, either alone or in conjunction with other WD **4110** components, such as device readable medium **4130**, WD **4110** functionality. Such functionality may include providing any of the various wireless features or benefits discussed herein. For example, processing circuitry **4120** may execute instructions stored in device readable medium **4130** or in memory within processing circuitry **4120** to provide the functionality disclosed herein.

As illustrated, processing circuitry **4120** includes one or more of RF transceiver circuitry **4122**, baseband processing circuitry **4124**, and application processing circuitry **4126**. In other embodiments, the processing circuitry may comprise

22

different components and/or different combinations of components. In certain embodiments processing circuitry **4120** of WD **4110** may comprise a SOC. In some embodiments, RF transceiver circuitry **4122**, baseband processing circuitry **4124**, and application processing circuitry **4126** may be on separate chips or sets of chips. In alternative embodiments, part or all of baseband processing circuitry **4124** and application processing circuitry **4126** may be combined into one chip or set of chips, and RF transceiver circuitry **4122** may be on a separate chip or set of chips. In still alternative embodiments, part or all of RF transceiver circuitry **4122** and baseband processing circuitry **4124** may be on the same chip or set of chips, and application processing circuitry **4126** may be on a separate chip or set of chips. In yet other alternative embodiments, part or all of RF transceiver circuitry **4122**, baseband processing circuitry **4124**, and application processing circuitry **4126** may be combined in the same chip or set of chips. In some embodiments, RF transceiver circuitry **4122** may be a part of interface **4114**. RF transceiver circuitry **4122** may condition RF signals for processing circuitry **4120**.

In certain embodiments, some or all of the functionality described herein as being performed by a WD may be provided by processing circuitry **4120** executing instructions stored on device readable medium **4130**, which in certain embodiments may be a computer-readable storage medium. In alternative embodiments, some or all of the functionality may be provided by processing circuitry **4120** without executing instructions stored on a separate or discrete device readable storage medium, such as in a hard-wired manner. In any of those particular embodiments, whether executing instructions stored on a device readable storage medium or not, processing circuitry **4120** can be configured to perform the described functionality. The benefits provided by such functionality are not limited to processing circuitry **4120** alone or to other components of WD **4110**, but are enjoyed by WD **4110** as a whole, and/or by end users and the wireless network generally.

Processing circuitry **4120** may be configured to perform any determining, calculating, or similar operations (e.g., certain obtaining operations) described herein as being performed by a WD. These operations, as performed by processing circuitry **4120**, may include processing information obtained by processing circuitry **4120** by, for example, converting the obtained information into other information, comparing the obtained information or converted information to information stored by WD **4110**, and/or performing one or more operations based on the obtained information or converted information, and as a result of said processing making a determination.

Device readable medium **4130** may be operable to store a computer program, software, an application including one or more of logic, rules, code, tables, etc. and/or other instructions capable of being executed by processing circuitry **4120**. Device readable medium **4130** may include computer memory (e.g., Random Access Memory (RAM) or Read Only Memory (ROM)), mass storage media (e.g., a hard disk), removable storage media (e.g., a Compact Disk (CD) or a Digital Video Disk (DVD)), and/or any other volatile or non-volatile, non-transitory device readable and/or computer executable memory devices that store information, data, and/or instructions that may be used by processing circuitry **4120**. In some embodiments, processing circuitry **4120** and device readable medium **4130** may be considered to be integrated.

User interface equipment **4132** may provide components that allow for a human user to interact with WD **4110**. Such

interaction may be of many forms, such as visual, audial, tactile, etc. User interface equipment **4132** may be operable to produce output to the user and to allow the user to provide input to WD **4110**. The type of interaction may vary depending on the type of user interface equipment **4132** installed in WD **4110**. For example, if WD **4110** is a smart phone, the interaction may be via a touch screen; if WD **4110** is a smart meter, the interaction may be through a screen that provides usage (e.g., the number of gallons used) or a speaker that provides an audible alert (e.g., if smoke is detected). User interface equipment **4132** may include input interfaces, devices and circuits, and output interfaces, devices and circuits. User interface equipment **4132** is configured to allow input of information into WD **4110**, and is connected to processing circuitry **4120** to allow processing circuitry **4120** to process the input information. User interface equipment **4132** may include, for example, a microphone, a proximity or other sensor, keys/buttons, a touch display, one or more cameras, a USB port, or other input circuitry. User interface equipment **4132** is also configured to allow output of information from WD **4110**, and to allow processing circuitry **4120** to output information from WD **4110**. User interface equipment **4132** may include, for example, a speaker, a display, vibrating circuitry, a USB port, a headphone interface, or other output circuitry. Using one or more input and output interfaces, devices, and circuits, of user interface equipment **4132**, WD **4110** may communicate with end users and/or the wireless network, and allow them to benefit from the functionality described herein.

Auxiliary equipment **4134** is operable to provide more specific functionality which may not be generally performed by WDs. This may comprise specialized sensors for doing measurements for various purposes, interfaces for additional types of communication such as wired communications etc. The inclusion and type of components of auxiliary equipment **4134** may vary depending on the embodiment and/or scenario.

Power source **4136** may, in some embodiments, be in the form of a battery or battery pack. Other types of power sources, such as an external power source (e.g., an electricity outlet), photovoltaic devices or power cells, may also be used. WD **4110** may further comprise power circuitry **4137** for delivering power from power source **4136** to the various parts of WD **4110** which need power from power source **4136** to carry out any functionality described or indicated herein. Power circuitry **4137** may in certain embodiments comprise power management circuitry. Power circuitry **4137** may additionally or alternatively be operable to receive power from an external power source; in which case WD **4110** may be connectable to the external power source (such as an electricity outlet) via input circuitry or an interface such as an electrical power cable. Power circuitry **4137** may also in certain embodiments be operable to deliver power from an external power source to power source **4136**. This may be, for example, for the charging of power source **4136**. Power circuitry **4137** may perform any formatting, converting, or other modification to the power from power source **4136** to make the power suitable for the respective components of WD **4110** to which power is supplied.

FIG. **10** illustrates a user Equipment in accordance with some embodiments.

FIG. **10** illustrates one embodiment of a UE in accordance with various aspects described herein. As used herein, a user equipment or UE may not necessarily have a user in the sense of a human user who owns and/or operates the relevant device. Instead, a UE may represent a device that is intended for sale to, or operation by, a human user but which may not,

or which may not initially, be associated with a specific human user (e.g., a smart sprinkler controller). Alternatively, a UE may represent a device that is not intended for sale to, or operation by, an end user but which may be associated with or operated for the benefit of a user (e.g., a smart power meter). UE **42200** may be any UE identified by the 3rd Generation Partnership Project (3GPP), including a NB-IoT UE, a machine type communication (MTC) UE, and/or an enhanced MTC (eMTC) UE. UE **4200**, as illustrated in FIG. **10**, is one example of a WD configured for communication in accordance with one or more communication standards promulgated by the 3rd Generation Partnership Project (3GPP), such as 3GPP's GSM, UMTS, LTE, and/or 5G standards. As mentioned previously, the term WD and UE may be used interchangeable. Accordingly, although FIG. **10** is a UE, the components discussed herein are equally applicable to a WD, and vice-versa.

In FIG. **10**, UE **4200** includes processing circuitry **4201** that is operatively coupled to input/output interface **4205**, radio frequency (RF) interface **4209**, network connection interface **4211**, memory **4215** including random access memory (RAM) **4217**, read-only memory (ROM) **4219**, and storage medium **4221** or the like, communication subsystem **4231**, power source **4213**, and/or any other component, or any combination thereof. Storage medium **4221** includes operating system **4223**, application program **4225**, and data **4227**. In other embodiments, storage medium **4221** may include other similar types of information. Certain UEs may utilize all of the components shown in FIG. **10**, or only a subset of the components. The level of integration between the components may vary from one UE to another UE. Further, certain UEs may contain multiple instances of a component, such as multiple processors, memories, transceivers, transmitters, receivers, etc.

In FIG. **10**, processing circuitry **4201** may be configured to process computer instructions and data. Processing circuitry **4201** may be configured to implement any sequential state machine operative to execute machine instructions stored as machine-readable computer programs in the memory, such as one or more hardware-implemented state machines (e.g., in discrete logic, FPGA, ASIC, etc.); programmable logic together with appropriate firmware; one or more stored program, general-purpose processors, such as a microprocessor or Digital Signal Processor (DSP), together with appropriate software; or any combination of the above. For example, the processing circuitry **4201** may include two central processing units (CPUs). Data may be information in a form suitable for use by a computer.

In the depicted embodiment, input/output interface **4205** may be configured to provide a communication interface to an input device, output device, or input and output device. UE **4200** may be configured to use an output device via input/output interface **4205**. An output device may use the same type of interface port as an input device. For example, a USB port may be used to provide input to and output from UE **4200**. The output device may be a speaker, a sound card, a video card, a display, a monitor, a printer, an actuator, an emitter, a smartcard, another output device, or any combination thereof. UE **4200** may be configured to use an input device via input/output interface **4205** to allow a user to capture information into UE **4200**. The input device may include a touch-sensitive or presence-sensitive display, a camera (e.g., a digital camera, a digital video camera, a web camera, etc.), a microphone, a sensor, a mouse, a trackball, a directional pad, a trackpad, a scroll wheel, a smartcard, and the like. The presence-sensitive display may include a capacitive or resistive touch sensor to sense input from a

25

user. A sensor may be, for instance, an accelerometer, a gyroscope, a tilt sensor, a force sensor, a magnetometer, an optical sensor, a proximity sensor, another like sensor, or any combination thereof. For example, the input device may be an accelerometer, a magnetometer, a digital camera, a microphone, and an optical sensor.

In FIG. 10, RF interface 4209 may be configured to provide a communication interface to RF components such as a transmitter, a receiver, and an antenna. Network connection interface 4211 may be configured to provide a communication interface to network 4243a. Network 4243a may encompass wired and/or wireless networks such as a local-area network (LAN), a wide-area network (WAN), a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network 4243a may comprise a Wi-Fi network. Network connection interface 4211 may be configured to include a receiver and a transmitter interface used to communicate with one or more other devices over a communication network according to one or more communication protocols, such as Ethernet, TCP/IP, SONET, ATM, or the like. Network connection interface 4211 may implement receiver and transmitter functionality appropriate to the communication network links (e.g., optical, electrical, and the like). The transmitter and receiver functions may share circuit components, software or firmware, or alternatively may be implemented separately.

RAM 4217 may be configured to interface via bus 4202 to processing circuitry 4201 to provide storage or caching of data or computer instructions during the execution of software programs such as the operating system, application programs, and device drivers. ROM 4219 may be configured to provide computer instructions or data to processing circuitry 4201. For example, ROM 4219 may be configured to store invariant low-level system code or data for basic system functions such as basic input and output (I/O), startup, or reception of keystrokes from a keyboard that are stored in a non-volatile memory. Storage medium 4221 may be configured to include memory such as RAM, ROM, programmable read-only memory (PROM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), magnetic disks, optical disks, floppy disks, hard disks, removable cartridges, or flash drives. In one example, storage medium 4221 may be configured to include operating system 4223, application program 4225 such as a web browser application, a widget or gadget engine or another application, and data file 4227. Storage medium 4221 may store, for use by UE 4200, any of a variety of various operating systems or combinations of operating systems.

Storage medium 4221 may be configured to include a number of physical drive units, such as redundant array of independent disks (RAID), floppy disk drive, flash memory, USB flash drive, external hard disk drive, thumb drive, pen drive, key drive, high-density digital versatile disc (HDDVD) optical disc drive, internal hard disk drive, Blu-Ray optical disc drive, holographic digital data storage (HDDS) optical disc drive, external mini-dual in-line memory module (DIMM), synchronous dynamic random access memory (SDRAM), external micro-DIMM SDRAM, smartcard memory such as a subscriber identity module or a removable user identity (SIM/RUIM) module, other memory, or any combination thereof. Storage medium 4221 may allow UE 4200 to access computer-executable instructions, application programs or the like, stored on transitory or non-transitory memory media, to off-load data, or to upload data. An article of manufacture, such as one utilizing a commu-

26

nication system may be tangibly embodied in storage medium 4221, which may comprise a device readable medium.

In FIG. 10, processing circuitry 4201 may be configured to communicate with network 4243b using communication subsystem 4231. Network 4243a and network 4243b may be the same network or networks or different network or networks. Communication subsystem 4231 may be configured to include one or more transceivers used to communicate with network 4243b. For example, communication subsystem 4231 may be configured to include one or more transceivers used to communicate with one or more remote transceivers of another device capable of wireless communication such as another WD, UE, or base station of a radio access network (RAN) according to one or more communication protocols, such as IEEE 802.11, CDMA, WCDMA, GSM, LTE, UTRAN, WiMax, or the like. Each transceiver may include transmitter 4233 and/or receiver 4235 to implement transmitter or receiver functionality, respectively, appropriate to the RAN links (e.g., frequency allocations and the like). Further, transmitter 4233 and receiver 4235 of each transceiver may share circuit components, software or firmware, or alternatively may be implemented separately.

In the illustrated embodiment, the communication functions of communication subsystem 4231 may include data communication, voice communication, multimedia communication, short-range communications such as Bluetooth, near-field communication, location-based communication such as the use of the global positioning system (GPS) to determine a location, another like communication function, or any combination thereof. For example, communication subsystem 4231 may include cellular communication, Wi-Fi communication, Bluetooth communication, and GPS communication. Network 4243b may encompass wired and/or wireless networks such as a local-area network (LAN), a wide-area network (WAN), a computer network, a wireless network, a telecommunications network, another like network or any combination thereof. For example, network 4243b may be a cellular network, a Wi-Fi network, and/or a near-field network. Power source 4213 may be configured to provide alternating current (AC) or direct current (DC) power to components of UE 4200.

The features, benefits and/or functions described herein may be implemented in one of the components of UE 4200 or partitioned across multiple components of UE 4200. Further, the features, benefits, and/or functions described herein may be implemented in any combination of hardware, software or firmware. In one example, communication subsystem 4231 may be configured to include any of the components described herein. Further, processing circuitry 4201 may be configured to communicate with any of such components over bus 4202. In another example, any of such components may be represented by program instructions stored in memory that when executed by processing circuitry 4201 perform the corresponding functions described herein. In another example, the functionality of any of such components may be partitioned between processing circuitry 4201 and communication subsystem 4231. In another example, the non-computationally intensive functions of any of such components may be implemented in software or firmware and the computationally intensive functions may be implemented in hardware.

FIG. 11 illustrates a virtualization environment in accordance with some embodiments.

FIG. 11 is a schematic block diagram illustrating a virtualization environment 4300 in which functions implemented by some embodiments may be virtualized. In the

present context, virtualizing means creating virtual versions of apparatuses or devices which may include virtualizing hardware platforms, storage devices and networking resources. As used herein, virtualization can be applied to a node (e.g., a virtualized base station or a virtualized radio access node) or to a device (e.g., a UE, a wireless device or any other type of communication device) or components thereof and relates to an implementation in which at least a portion of the functionality is implemented as one or more virtual components (e.g., via one or more applications, components, functions, virtual machines or containers executing on one or more physical processing nodes in one or more networks).

In some embodiments, some or all of the functions described herein may be implemented as virtual components executed by one or more virtual machines implemented in one or more virtual environments **4300** hosted by one or more of hardware nodes **4330**. Further, in embodiments in which the virtual node is not a radio access node or does not require radio connectivity (e.g., a core network node), then the network node may be entirely virtualized.

The functions may be implemented by one or more applications **4320** (which may alternatively be called software instances, virtual appliances, network functions, virtual nodes, virtual network functions, etc.) operative to implement some of the features, functions, and/or benefits of some of the embodiments disclosed herein. Applications **4320** are run in virtualization environment **4300** which provides hardware **4330** comprising processing circuitry **4360** and memory **4390**. Memory **4390** contains instructions **4395** executable by processing circuitry **4360** whereby application **4320** is operative to provide one or more of the features, benefits, and/or functions disclosed herein.

Virtualization environment **4300**, comprises general-purpose or special-purpose network hardware devices **4330** comprising a set of one or more processors or processing circuitry **4360**, which may be commercial off-the-shelf (COTS) processors, dedicated Application Specific Integrated Circuits (ASICs), or any other type of processing circuitry including digital or analog hardware components or special purpose processors. Each hardware device may comprise memory **4390-1** which may be non-persistent memory for temporarily storing instructions **4395** or software executed by processing circuitry **4360**. Each hardware device may comprise one or more network interface controllers (NICs) **4370**, also known as network interface cards, which include physical network interface **4380**. Each hardware device may also include non-transitory, persistent, machine-readable storage media **4390-2** having stored therein software **4395** and/or instructions executable by processing circuitry **4360**. Software **4395** may include any type of software including software for instantiating one or more virtualization layers **4350** (also referred to as hypervisors), software to execute virtual machines **4340** as well as software allowing it to execute functions, features and/or benefits described in relation with some embodiments described herein.

Virtual machines **4340** comprise virtual processing, virtual memory, virtual networking or interface and virtual storage, and may be run by a corresponding virtualization layer **4350** or hypervisor. Different embodiments of the instance of virtual appliance **4320** may be implemented on one or more of virtual machines **4340**, and the implementations may be made in different ways.

During operation, processing circuitry **4360** executes software **4395** to instantiate the hypervisor or virtualization layer **4350**, which may sometimes be referred to as a virtual

machine monitor (VMM). Virtualization layer **4350** may present a virtual operating platform that appears like networking hardware to virtual machine **4340**.

As shown in FIG. 11, hardware **4330** may be a standalone network node with generic or specific components. Hardware **4330** may comprise antenna **43225** and may implement some functions via virtualization. Alternatively, hardware **4330** may be part of a larger cluster of hardware (e.g. such as in a data center or customer premise equipment (CPE)) where many hardware nodes work together and are managed via management and orchestration (MANO) **43100**, which, among others, oversees lifecycle management of applications **4320**.

Virtualization of the hardware is in some contexts referred to as network function virtualization (NFV). NFV may be used to consolidate many network equipment types onto industry standard high volume server hardware, physical switches, and physical storage, which can be located in data centers, and customer premise equipment.

In the context of NFV, virtual machine **4340** may be a software implementation of a physical machine that runs programs as if they were executing on a physical, non-virtualized machine. Each of virtual machines **4340**, and that part of hardware **4330** that executes that virtual machine, be it hardware dedicated to that virtual machine and/or hardware shared by that virtual machine with others of the virtual machines **4340**, forms a separate virtual network elements (VNE).

Still in the context of NFV, Virtual Network Function (VNF) is responsible for handling specific network functions that run in one or more virtual machines **4340** on top of hardware networking infrastructure **4330** and corresponds to application **4320** in FIG. 11.

In some embodiments, one or more radio units **43200** that each include one or more transmitters **43220** and one or more receivers **43210** may be coupled to one or more antennas **43225**. Radio units **43200** may communicate directly with hardware nodes **4330** via one or more appropriate network interfaces and may be used in combination with the virtual components to provide a virtual node with radio capabilities, such as a radio access node or a base station.

In some embodiments, some signalling can be effected with the use of control system **43230** which may alternatively be used for communication between the hardware nodes **4330** and radio units **43200**.

FIG. 12 illustrates a telecommunication network connected via an intermediate network to a host computer in accordance with some embodiments.

With reference to FIG. 12, in accordance with an embodiment, a communication system includes telecommunication network **4410**, such as a 3GPP-type cellular network, which comprises access network **4411**, such as a radio access network, and core network **4414**. Access network **4411** comprises a plurality of base stations **4412a**, **4412b**, **4412c**, such as NBs, eNBs, gNBs or other types of wireless access points, each defining a corresponding coverage area **4413a**, **4413b**, **4413c**. Each base station **4412a**, **4412b**, **4412c** is connectable to core network **4414** over a wired or wireless connection **4415**. A first UE **4491** located in coverage area **4413c** is configured to wirelessly connect to, or be paged by, the corresponding base station **4412c**. A second UE **4492** in coverage area **4413a** is wirelessly connectable to the corresponding base station **4412a**. While a plurality of UEs **4491**, **4492** are illustrated in this example, the disclosed embodiments are equally applicable to a situation where a sole UE

is in the coverage area or where a sole UE is connecting to the corresponding base station **4412**.

Telecommunication network **4410** is itself connected to host computer **4430**, which may be embodied in the hardware and/or software of a standalone server, a cloud-implemented server, a distributed server or as processing resources in a server farm. Host computer **4430** may be under the ownership or control of a service provider, or may be operated by the service provider or on behalf of the service provider. Connections **4421** and **4422** between telecommunication network **4410** and host computer **4430** may extend directly from core network **4414** to host computer **4430** or may go via an optional intermediate network **4420**. Intermediate network **4420** may be one of, or a combination of more than one of, a public, private or hosted network; intermediate network **4420**, if any, may be a backbone network or the Internet; in particular, intermediate network **4420** may comprise two or more sub-networks (not shown).

The communication system of FIG. **12** as a whole enables connectivity between the connected UEs **4491**, **4492** and host computer **4430**. The connectivity may be described as an over-the-top (OTT) connection **4450**. Host computer **4430** and the connected UEs **4491**, **4492** are configured to communicate data and/or signaling via OTT connection **4450**, using access network **4411**, core network **4414**, any intermediate network **4420** and possible further infrastructure (not shown) as intermediaries. OTT connection **4450** may be transparent in the sense that the participating communication devices through which OTT connection **4450** passes are unaware of routing of uplink and downlink communications. For example, base station **4412** may not or need not be informed about the past routing of an incoming downlink communication with data originating from host computer **4430** to be forwarded (e.g., handed over) to a connected UE **4491**. Similarly, base station **4412** need not be aware of the future routing of an outgoing uplink communication originating from the UE **4491** towards the host computer **4430**.

FIG. **13** illustrates a host computer communicating via a base station with a user equipment over a partially wireless connection in accordance with some embodiments.

Example implementations, in accordance with an embodiment, of the UE, base station and host computer discussed in the preceding paragraphs will now be described with reference to FIG. **13**. In communication system **4500**, host computer **4510** comprises hardware **4515** including communication interface **4516** configured to set up and maintain a wired or wireless connection with an interface of a different communication device of communication system **4500**. Host computer **4510** further comprises processing circuitry **4518**, which may have storage and/or processing capabilities. In particular, processing circuitry **4518** may comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions. Host computer **4510** further comprises software **4511**, which is stored in or accessible by host computer **4510** and executable by processing circuitry **4518**. Software **4511** includes host application **4512**. Host application **4512** may be operable to provide a service to a remote user, such as UE **4530** connecting via OTT connection **4550** terminating at UE **4530** and host computer **4510**. In providing the service to the remote user, host application **4512** may provide user data which is transmitted using OTT connection **4550**.

Communication system **4500** further includes base station **4520** provided in a telecommunication system and compris-

ing hardware **4525** enabling it to communicate with host computer **4510** and with UE **4530**. Hardware **4525** may include communication interface **4526** for setting up and maintaining a wired or wireless connection with an interface of a different communication device of communication system **4500**, as well as radio interface **4527** for setting up and maintaining at least wireless connection **4570** with UE **4530** located in a coverage area (not shown in FIG. **13**) served by base station **4520**. Communication interface **4526** may be configured to facilitate connection **4560** to host computer **4510**. Connection **4560** may be direct or it may pass through a core network (not shown in FIG. **13**) of the telecommunication system and/or through one or more intermediate networks outside the telecommunication system. In the embodiment shown, hardware **4525** of base station **4520** further includes processing circuitry **4528**, which may comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions. Base station **4520** further has software **4521** stored internally or accessible via an external connection.

Communication system **4500** further includes UE **4530** already referred to. Its hardware **4535** may include radio interface **4537** configured to set up and maintain wireless connection **4570** with a base station serving a coverage area in which UE **4530** is currently located. Hardware **4535** of UE **4530** further includes processing circuitry **4538**, which may comprise one or more programmable processors, application-specific integrated circuits, field programmable gate arrays or combinations of these (not shown) adapted to execute instructions. UE **4530** further comprises software **4531**, which is stored in or accessible by UE **4530** and executable by processing circuitry **4538**. Software **4531** includes client application **4532**. Client application **4532** may be operable to provide a service to a human or non-human user via UE **4530**, with the support of host computer **4510**. In host computer **4510**, an executing host application **4512** may communicate with the executing client application **4532** via OTT connection **4550** terminating at UE **4530** and host computer **4510**. In providing the service to the user, client application **4532** may receive request data from host application **4512** and provide user data in response to the request data. OTT connection **4550** may transfer both the request data and the user data. Client application **4532** may interact with the user to generate the user data that it provides.

It is noted that host computer **4510**, base station **4520** and UE **4530** illustrated in FIG. **13** may be similar or identical to host computer **4430**, one of base stations **4412a**, **4412b**, **4412c** and one of UEs **4491**, **4492** of FIG. **12**, respectively. This is to say, the inner workings of these entities may be as shown in FIG. **13** and independently, the surrounding network topology may be that of FIG. **12**.

In FIG. **13**, OTT connection **4550** has been drawn abstractly to illustrate the communication between host computer **4510** and UE **4530** via base station **4520**, without explicit reference to any intermediary devices and the precise routing of messages via these devices. Network infrastructure may determine the routing, which it may be configured to hide from UE **4530** or from the service provider operating host computer **4510**, or both. While OTT connection **4550** is active, the network infrastructure may further take decisions by which it dynamically changes the routing (e.g., on the basis of load balancing consideration or reconfiguration of the network).

31

Wireless connection **4570** between UE **4530** and base station **4520** is in accordance with the teachings of the embodiments described throughout this disclosure. One or more of the various embodiments may improve the performance of OTT services provided to UE **4530** using OTT connection **4550**, in which wireless connection **4570** forms the last segment. More precisely, the teachings of these embodiments may improve the random access speed and/or reduce random access failure rates and thereby provide benefits such as faster and/or more reliable random access.

A measurement procedure may be provided for the purpose of monitoring data rate, latency and other factors on which the one or more embodiments improve. There may further be an optional network functionality for reconfiguring OTT connection **4550** between host computer **4510** and UE **4530**, in response to variations in the measurement results. The measurement procedure and/or the network functionality for reconfiguring OTT connection **4550** may be implemented in software **4511** and hardware **4515** of host computer **4510** or in software **4531** and hardware **4535** of UE **4530**, or both. In embodiments, sensors (not shown) may be deployed in or in association with communication devices through which OTT connection **4550** passes; the sensors may participate in the measurement procedure by supplying values of the monitored quantities exemplified above, or supplying values of other physical quantities from which software **4511**, **4531** may compute or estimate the monitored quantities. The reconfiguring of OTT connection **4550** may include message format, retransmission settings, preferred routing etc.; the reconfiguring need not affect base station **4520**, and it may be unknown or imperceptible to base station **4520**. Such procedures and functionalities may be known and practiced in the art. In certain embodiments, measurements may involve proprietary UE signaling facilitating host computer **4510**'s measurements of throughput, propagation times, latency and the like. The measurements may be implemented in that software **4511** and **4531** causes messages to be transmitted, in particular empty or 'dummy' messages, using OTT connection **4550** while it monitors propagation times, errors etc.

FIG. **14** illustrates methods implemented in a communication system including a host computer, a base station and a user equipment in accordance with some embodiments

FIG. **14** is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS. **12** and **13**. For simplicity of the present disclosure, only drawing references to FIG. **14** will be included in this section. In step **4610**, the host computer provides user data. In substep **4611** (which may be optional) of step **4610**, the host computer provides the user data by executing a host application. In step **4620**, the host computer initiates a transmission carrying the user data to the UE. In step **4630** (which may be optional), the base station transmits to the UE the user data which was carried in the transmission that the host computer initiated, in accordance with the teachings of the embodiments described throughout this disclosure. In step **4640** (which may also be optional), the UE executes a client application associated with the host application executed by the host computer.

FIG. **15** illustrates methods implemented in a communication system including a host computer, a base station and a user equipment in accordance with some embodiments.

FIG. **15** is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer,

32

a base station and a UE which may be those described with reference to FIGS. **12** and **13**. For simplicity of the present disclosure, only drawing references to FIG. **15** will be included in this section. In step **4710** of the method, the host computer provides user data. In an optional substep (not shown) the host computer provides the user data by executing a host application. In step **4720**, the host computer initiates a transmission carrying the user data to the UE. The transmission may pass via the base station, in accordance with the teachings of the embodiments described throughout this disclosure. In step **4730** (which may be optional), the UE receives the user data carried in the transmission.

FIG. **16** illustrates methods implemented in a communication system including a host computer, a base station and a user equipment in accordance with some embodiments

FIG. **16** is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS. **12** and **13**. For simplicity of the present disclosure, only drawing references to FIG. **16** will be included in this section. In step **4810** (which may be optional), the UE receives input data provided by the host computer. Additionally or alternatively, in step **4820**, the UE provides user data. In substep **4821** (which may be optional) of step **4820**, the UE provides the user data by executing a client application. In substep **4811** (which may be optional) of step **4810**, the UE executes a client application which provides the user data in reaction to the received input data provided by the host computer. In providing the user data, the executed client application may further consider user input received from the user. Regardless of the specific manner in which the user data was provided, the UE initiates, in substep **4830** (which may be optional), transmission of the user data to the host computer. In step **4840** of the method, the host computer receives the user data transmitted from the UE, in accordance with the teachings of the embodiments described throughout this disclosure.

FIG. **17** illustrates methods implemented in a communication system including a host computer, a base station and a user equipment in accordance with some embodiments

FIG. **17** is a flowchart illustrating a method implemented in a communication system, in accordance with one embodiment. The communication system includes a host computer, a base station and a UE which may be those described with reference to FIGS. **12** and **13**. For simplicity of the present disclosure, only drawing references to FIG. **17** will be included in this section. In step **4910** (which may be optional), in accordance with the teachings of the embodiments described throughout this disclosure, the base station receives user data from the UE. In step **4920** (which may be optional), the base station initiates transmission of the received user data to the host computer. In step **4930** (which may be optional), the host computer receives the user data carried in the transmission initiated by the base station.

Any appropriate steps, methods, features, functions, or benefits disclosed herein may be performed through one or more functional units or modules of one or more virtual apparatuses. Each virtual apparatus may comprise a number of these functional units. These functional units may be implemented via processing circuitry, which may include one or more microprocessor or microcontrollers, as well as other digital hardware, which may include digital signal processors (DSPs), special-purpose digital logic, and the like. The processing circuitry may be configured to execute program code stored in memory, which may include one or several types of memory such as read-only memory (ROM),

33

random-access memory (RAM), cache memory, flash memory devices, optical storage devices, etc. Program code stored in memory includes program instructions for executing one or more telecommunications and/or data communications protocols as well as instructions for carrying out one or more of the techniques described herein. In some implementations, the processing circuitry may be used to cause the respective functional unit to perform corresponding functions according one or more embodiments of the present disclosure.

The term unit may have conventional meaning in the field of electronics, electrical devices and/or electronic devices and may include, for example, electrical and/or electronic circuitry, devices, modules, processors, memories, logic solid state and/or discrete devices, computer programs or instructions for carrying out respective tasks, procedures, computations, outputs, and/or displaying functions, and so on, as such as those that are described herein.

ABBREVIATIONS

At least some of the following abbreviations may be used in this disclosure. If there is an inconsistency between abbreviations, preference should be given to how it is used above. If listed multiple times below, the first listing should be preferred over any subsequent listing(s).

1×RTT CDMA2000 1× Radio Transmission Technology
 3GPP 3rd Generation Partnership Project
 5G 5th Generation
 ABS Almost Blank Subframe
 ARQ Automatic Repeat Request
 AWGN Additive White Gaussian Noise
 BCCH Broadcast Control Channel
 BCH Broadcast Channel
 CA Carrier Aggregation
 CC Carrier Component
 CCCH SDU Common Control Channel SDU
 CDMA Code Division Multiplexing Access
 CGI Cell Global Identifier
 CIR Channel Impulse Response
 CP Cyclic Prefix
 CPICH Common Pilot Channel
 CPICH Ec/No CPICH Received energy per chip divided by the power density in the band
 CQI Channel Quality information
 C-RNTI Cell RNTI
 CSI Channel State Information
 DCCH Dedicated Control Channel
 DL Downlink
 DM Demodulation
 DMRS Demodulation Reference Signal
 DRX Discontinuous Reception
 DTX Discontinuous Transmission
 DTCH Dedicated Traffic Channel
 DUT Device Under Test
 E-CID Enhanced Cell-ID (positioning method)
 E-SMLC Evolved-Serving Mobile Location Centre
 ECGI Evolved CGI
 eNB E-UTRAN NodeB
 ePDCCH enhanced Physical Downlink Control Channel
 E-SMLC evolved Serving Mobile Location Center
 E-UTRA Evolved UTRA
 E-UTRAN Evolved UTRAN
 FDD Frequency Division Duplex
 FFS For Further Study
 GERAN GSM EDGE Radio Access Network
 gNB Base station in NR

34

GNSS Global Navigation Satellite System
 GSM Global System for Mobile communication
 HARQ Hybrid Automatic Repeat Request
 HO Handover
 HSPA High Speed Packet Access
 HRPD High Rate Packet Data
 LOS Line of Sight
 LPP LTE Positioning Protocol
 LTE Long-Term Evolution
 MAC Medium Access Control
 MBMS Multimedia Broadcast Multicast Services
 MBSFN Multimedia Broadcast multicast service Single Frequency Network
 MBSFN ABS MBSFN Almost Blank Subframe
 MDT Minimization of Drive Tests
 MIB Master Information Block
 MME Mobility Management Entity
 MSC Mobile Switching Center
 NPDCCH Narrowband Physical Downlink Control Channel
 NR New Radio
 OCNF OFDMA Channel Noise Generator
 OFDM Orthogonal Frequency Division Multiplexing
 OFDMA Orthogonal Frequency Division Multiple Access
 OSS Operations Support System
 OTDOA Observed Time Difference of Arrival
 O&M Operation and Maintenance
 PBCH Physical Broadcast Channel
 P-CCPCH Primary Common Control Physical Channel
 PCell Primary Cell
 PCFICH Physical Control Format Indicator Channel
 PDCCH Physical Downlink Control Channel
 PDP Profile Delay Profile
 PDSCH Physical Downlink Shared Channel
 PGW Packet Gateway
 PHICH Physical Hybrid-ARQ Indicator Channel
 PLMN Public Land Mobile Network
 PMI Precoder Matrix Indicator
 PRACH Physical Random Access Channel
 PRS Positioning Reference Signal
 PSS Primary Synchronization Signal
 PUCCH Physical Uplink Control Channel
 PUSCH Physical Uplink Shared Channel
 RACH Random Access Channel
 QAM Quadrature Amplitude Modulation
 RAN Radio Access Network
 RAT Radio Access Technology
 RLM Radio Link Management
 RNC Radio Network Controller
 RNTI Radio Network Temporary Identifier
 RRC Radio Resource Control
 RRM Radio Resource Management
 RS Reference Signal
 RSCP Received Signal Code Power
 RSRP Reference Symbol Received Power OR Reference Signal Received Power
 RSRQ Reference Signal Received Quality OR Reference Symbol Received Quality
 RSSI Received Signal Strength Indicator
 RSTD Reference Signal Time Difference
 SCH Synchronization Channel
 SCell Secondary Cell
 SDU Service Data Unit
 SFN System Frame Number
 SGW Serving Gateway
 SI System Information
 SIB System Information Block

35

SNR Signal to Noise Ratio
 SON Self Optimized Network
 SS Synchronization Signal
 SSS Secondary Synchronization Signal
 TDD Time Division Duplex
 TDOA Time Difference of Arrival
 TOA Time of Arrival
 TSS Tertiary Synchronization Signal
 TTI Transmission Time Interval
 UE User Equipment
 UL Uplink
 UMTS Universal Mobile Telecommunication System
 USIM Universal Subscriber Identity Module
 UTDOA Uplink Time Difference of Arrival
 UTRA Universal Terrestrial Radio Access
 UTRAN Universal Terrestrial Radio Access Network
 WCDMA Wide CDMA
 WLAN Wide Local Area Network

Further definitions and embodiments are discussed below.

In the above-description of various embodiments of present inventive concepts, it is to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of present inventive concepts. Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which present inventive concepts belong. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

When an element is referred to as being “connected”, “coupled”, “responsive”, or variants thereof to another element, it can be directly connected, coupled, or responsive to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected”, “directly coupled”, “directly responsive”, or variants thereof to another element, there are no intervening elements present. Like numbers refer to like elements throughout. Furthermore, “coupled”, “connected”, “responsive”, or variants thereof as used herein may include wirelessly coupled, connected, or responsive. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Well-known functions or constructions may not be described in detail for brevity and/or clarity. The term “and/or” (abbreviated “/”) includes any and all combinations of one or more of the associated listed items.

It will be understood that although the terms first, second, third, etc. may be used herein to describe various elements/operations, these elements/operations should not be limited by these terms. These terms are only used to distinguish one element/operation from another element/operation. Thus a first element/operation in some embodiments could be termed a second element/operation in other embodiments without departing from the teachings of present inventive concepts. The same reference numerals or the same reference designators denote the same or similar elements throughout the specification.

As used herein, the terms “comprise”, “comprising”, “comprises”, “include”, “including”, “includes”, “have”, “has”, “having”, or variants thereof are open-ended, and include one or more stated features, integers, elements, steps, components or functions but does not preclude the presence or addition of one or more other features, integers,

36

elements, steps, components, functions or groups thereof. Furthermore, as used herein, the common abbreviation “e.g.”, which derives from the Latin phrase “*exempli gratia*,” may be used to introduce or specify a general example or examples of a previously mentioned item, and is not intended to be limiting of such item. The common abbreviation “i.e.”, which derives from the Latin phrase “*id est*,” may be used to specify a particular item from a more general recitation.

Example embodiments are described herein with reference to block diagrams and/or flowchart illustrations of computer-implemented methods, apparatus (systems and/or devices) and/or computer program products. It is understood that a block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by computer program instructions that are performed by one or more computer circuits. These computer program instructions may be provided to a processor circuit of a general purpose computer circuit, special purpose computer circuit, and/or other programmable data processing circuit to produce a machine, such that the instructions, which execute via the processor of the computer and/or other programmable data processing apparatus, transform and control transistors, values stored in memory locations, and other hardware components within such circuitry to implement the functions/acts specified in the block diagrams and/or flowchart block or blocks, and thereby create means (functionality) and/or structure for implementing the functions/acts specified in the block diagrams and/or flowchart block(s).

These computer program instructions may also be stored in a tangible computer-readable medium that can direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable medium produce an article of manufacture including instructions which implement the functions/acts specified in the block diagrams and/or flowchart block or blocks. Accordingly, embodiments of present inventive concepts may be embodied in hardware and/or in software (including firmware, resident software, microcode, etc.) that runs on a processor such as a digital signal processor, which may collectively be referred to as “circuitry,” “a module” or variants thereof.

It should also be noted that in some alternate implementations, the functions/acts noted in the blocks may occur out of the order noted in the flowcharts. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved. Moreover, the functionality of a given block of the flowcharts and/or block diagrams may be separated into multiple blocks and/or the functionality of two or more blocks of the flowcharts and/or block diagrams may be at least partially integrated. Finally, other blocks may be added/inserted between the blocks that are illustrated, and/or blocks/operations may be omitted without departing from the scope of inventive concepts. Moreover, although some of the diagrams include arrows on communication paths to show a primary direction of communication, it is to be understood that communication may occur in the opposite direction to the depicted arrows.

Many variations and modifications can be made to the embodiments without substantially departing from the principles of the present inventive concepts. All such variations and modifications are intended to be included herein within the scope of present inventive concepts. Accordingly, the above disclosed subject matter is to be considered illustrative

tive, and not restrictive, and the examples of embodiments are intended to cover all such modifications, enhancements, and other embodiments, which fall within the spirit and scope of present inventive concepts. Thus, to the maximum extent allowed by law, the scope of present inventive concepts are to be determined by the broadest permissible interpretation of the present disclosure including the examples of embodiments and their equivalents, and shall not be restricted or limited by the foregoing detailed description.

The invention claimed is:

1. A method for reporting hybrid automatic repeat request-acknowledgement, HARQ-ACK, code books of fixed sizes in a communication network in a user equipment, UE, communication device, the method comprising:

responsive to operating in a first mode, reporting a HARQ-ACK codebook for each scheduled transmission to a network node;

responsive to operating in a second mode, reporting the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the HARQ-ACK codebook is ACK, wherein no NACK is transmitted in the HARQ-ACK codebook when operating in the second mode;

responsive to operating in a third mode, reporting the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the HARQ-ACK codebook is negative acknowledgement, NACK, wherein no ACK is transmitted in the HARQ-ACK codebook when operating in the third mode;

a Type 3 HARQ codebook being used in combination with any of the first mode, the second mode, and the third mode;

extending the Type 3 HARQ codebook to support triggering for a subset of HARQ processes instead of all HARQ processes; and

determining a HARQ feedback to be reported by at least one of:

reporting ACK/NACK to the network node corresponding to a latest obtained feedback for semi-persistent scheduling, SPS, physical downlink shared channel, PDSCH, occasion for a certain HARQ process occurrence before a physical uplink control channel, PUCCH, occasion where Type 3 HARQ codebook should be sent; and

reporting ACK/NACK based on a number of SPS PDSCH occurrences within a past time window.

2. The method of claim 1, further comprising:

responsive to an SPS PDSCH, not be decoded/detected successfully, skip sending NACK in a PUCCH occasion corresponding to a SPS PDSCH reception.

3. The method of claim 1, further comprising:

configuring the UE communication device to periodically send, to the network node, HARQ-feedback report that includes feedback corresponding to an SPS configured HARQ processes.

4. The method of claim 1, further comprising:

responsive to operating in the first mode, transmitting, to the network node, an ACK responsive to correctly decoding PDSCH and transmitting, to the network node, a NACK for cases with no detection or misdetection, or if demodulation reference signal, DMRS, is detected but data is not;

responsive to operating in the second mode, transmitting, to the network node, the ACK responsive to correctly decoding PDSCH and not transmitting, to the network

node, a NACK for cases with no detection or misdetection, or if DMRS is detected but data is not;

responsive to operating in the third mode, not transmitting, to the network node, an ACK responsive to correctly decoding PDSCH and transmitting, to the network node, a NACK for cases with no detection or misdetection, or if DMRS is detected but data is not; and

responsive to operating in a fourth mode, transmitting no feedback to the network node.

5. The method of claim 1, wherein HARQ-feedback uplink control information, UCI, of any type is sent to the network node in a PUCCH or in a PUSCH.

6. The method of claim 1, further comprising:

responsive to a PUCCH carrying HARQ feedback corresponding to an SPS PDSCH reception colliding with a PUCCH carrying Type3 HARQ codebook, transmitting only the Type3 HARQ codebook to the network node.

7. The method of claim 1, wherein responsive to downlink control information, DCI, having priority configured, providing feedback to the network node for indicated priority SPSs' PDSCHs in the HARQ-ACK codebook.

8. The method of claim 1, wherein high priority downlink control information, DCI, asking one-shot based feedback overrides low priority DCI.

9. The method of claim 8 wherein the DCI asking one-shot based feedback is transmitted in PDSCHs.

10. The method of claim 1, further comprising arranging HARQ-bits in the HARQ-ACK codebook according to HARQ process or according to SPS indices and then according to HARQ processes per SPS index.

11. A communication device for reporting hybrid automatic repeat request-acknowledgement, HARQ-ACK, code books of fixed sizes in a communication network, the communication device comprising:

processing circuitry; and

memory coupled with the processing circuitry, wherein the memory includes instructions that when executed by the processing circuitry causes the communication device to:

responsive to operating in a first mode, report a HARQ-ACK codebook for each scheduled transmission to a network node;

responsive to operating in a second mode, report the HARQ-ACK codebook to the network node if at least one of HARQ-ACK information in the codebook is ACK, wherein no NACK is transmitted in the HARQ-ACK codebook when operating in the second mode;

responsive to operating in a third mode, report the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the HARQ-ACK codebook is negative acknowledgement, NACK, wherein no ACK is transmitted in the HARQ-ACK codebook when operating in the third mode;

a Type 3 HARQ codebook being used in combination with any of the first mode, the second mode, and the third mode;

extend the Type 3 HARQ codebook to support triggering for a subset of HARQ processes instead of all HARQ processes; and

determine a HARQ feedback to be reported by at least one of:

reporting ACK/NACK to the network node corresponding to a latest obtained feedback for semi-persistent scheduling, SPS, physical downlink

39

shared channel, PDSCH, occasion for a certain HARQ process occurrence before a physical uplink control channel, PUCCH, occasion where Type 3 HARQ codebook should be sent; and reporting ACK/NACK based on a number of SPS PDSCH occurrences within a past time window.

12. The communication device of claim 11, wherein the memory includes further instructions that when executed by the processing circuitry causes the communication device to: responsive to a SPS PDSCH not be decoded/detected successfully, skip sending NACK in the PUCCH occasion corresponding to a SPS PDSCH reception.

13. The communication device of claim 11, wherein the memory includes further instructions that when executed by the processing circuitry causes the communication device to: configure the communication device to periodically send, to the network node, HARQ-feedback report that includes feedback corresponding to an SPS configured HARQ processes.

14. The communication device of claim 11, wherein the memory includes further instructions that when executed by the processing circuitry causes the communication device to: responsive to operating in the first mode, transmit, to the network node, an ACK responsive to correctly decoding PDSCH and transmitting, to the network node, a NACK for cases with no detection or misdetection, or if DMRS is detected but data is not; responsive to operating in the second mode, transmit, to the network node, the ACK responsive to correctly decoding PDSCH and not transmitting, to the network node, a NACK for cases with no detection or misdetection, or if DMRS is detected but data is not; responsive to operating in a third mode, not transmit, to the network node, an ACK responsive to correctly decoding PDSCH and transmitting, to the network node, a NACK for cases with no detection or misdetection, or if DMRS is detected but data is not; and responsive to operating in a fourth mode, transmit no feedback to the network node.

15. The communication device of claim 11, wherein HARQ-feedback uplink control information, UCI, of any type is sent to the network node in a PUCCH or in a PUSCH.

16. The communication device of claim 11, wherein the memory includes instructions that when executed by the processing circuitry causes the communication device to: responsive to a PUCCH carrying HARQ feedback corresponding to an SPS PDSCH reception colliding with a

40

PUCCH carrying Type3 HARQ codebook, transmit only the Type3 HARQ codebook to the network node.

17. The communication device of claim 11, wherein the memory includes further instructions that when executed by the processing circuitry causes the communication device to: responsive to downlink control information, DCI, having priority configured, provide feedback to the network node for high priority SPSs' PDSCHs in the HARQ-ACK codebook.

18. A non-transitory computer storage medium storing a computer program comprising program code to be executed by processing circuitry of a communication device, execution of the program code causing the communication device to:

responsive to operating in a first mode, report a hybrid automatic repeat request-acknowledgement, HARQ-ACK, codebook for each scheduled transmission to a network node;

responsive to operating in a second mode, report the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the codebook is ACK, wherein no NACK is transmitted in the HARQ-ACK codebook when operating in the second mode;

responsive to operating in a third mode, report the HARQ-ACK codebook to the network node if at least one of the HARQ-ACK information in the HARQ-ACK codebook is negative acknowledgement, NACK, wherein no ACK is transmitted in the HARQ-ACK codebook when operating in the third mode;

a Type 3 HARQ codebook being used in combination with any of the first mode, the second mode, and the third mode;

extend the Type 3 HARQ codebook to support triggering for a subset of HARQ processes instead of all HARQ processes; and

determine a HARQ feedback to be reported by at least one of:

reporting ACK/NACK to the network node corresponding to a latest obtained feedback for semi-persistent scheduling, SPS, physical downlink shared channel, PDSCH, occasion for a certain HARQ process occurrence before a physical uplink control channel, PUCCH, occasion where Type 3 HARQ codebook should be sent; and

reporting ACK/NACK based on a number of SPS PDSCH occurrences within a past time window.

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